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(54) **THREE-DIMENSIONAL SHAPING DEVICE
WITH NOZZLE FLUSHING OPERATION**

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,660,621 A * 8/1997 Bredt B29C 41/003
106/287.34
2008/0006958 A1 * 1/2008 Davidson B33Y 40/00
425/72.1

(Continued)

FOREIGN PATENT DOCUMENTS

DE 102009055966 A1 * 6/2011 B29C 67/0081
JP 2008513252 A * 5/2008

(Continued)

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B22F 12/90 (2021.01)
B29C 64/165 (2017.01)
B29C 64/209 (2017.01)

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(2021.01); **B22F 12/90** (2021.01); **B29C**
64/165 (2017.08); **B29C 64/209** (2017.08);
B33Y 30/00 (2014.12); **B33Y 50/02** (2014.12)

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B29C 64/30; B22F 12/50; B22F 12/90;
B22F 12/53; B33Y 30/00; B33Y 50/02;
B33Y 40/00

See application file for complete search history.

OTHER PUBLICATIONS

Proxima 3D. Binder Jetting 3D Printing Process. Feb. 16, 2016.
Accessed from www.youtube.com/watch?v=97doBH9jSXY (Year:
2016).*

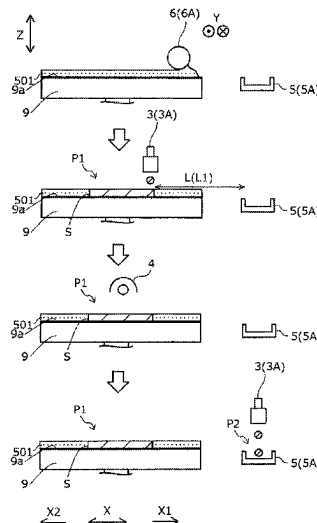
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(57) **ABSTRACT**

A three-dimensional shaping device includes: a shaping
table; a layer forming unit configured to form a powder layer
on the shaping table; a head configured to eject a liquid
containing a binder from a nozzle to a shaping region; and
a control unit configured to control movement of the head
with respect to the shaping table and driving of the head by
applying a voltage. The control unit changes an execution
condition of a flushing operation depending on whether an
idle running time during which the head is moved without
ejecting the liquid is less than a threshold value or equal to
or greater than the threshold value.

14 Claims, 15 Drawing Sheets



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B33Y 30/00 (2015.01)
B33Y 50/02 (2015.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

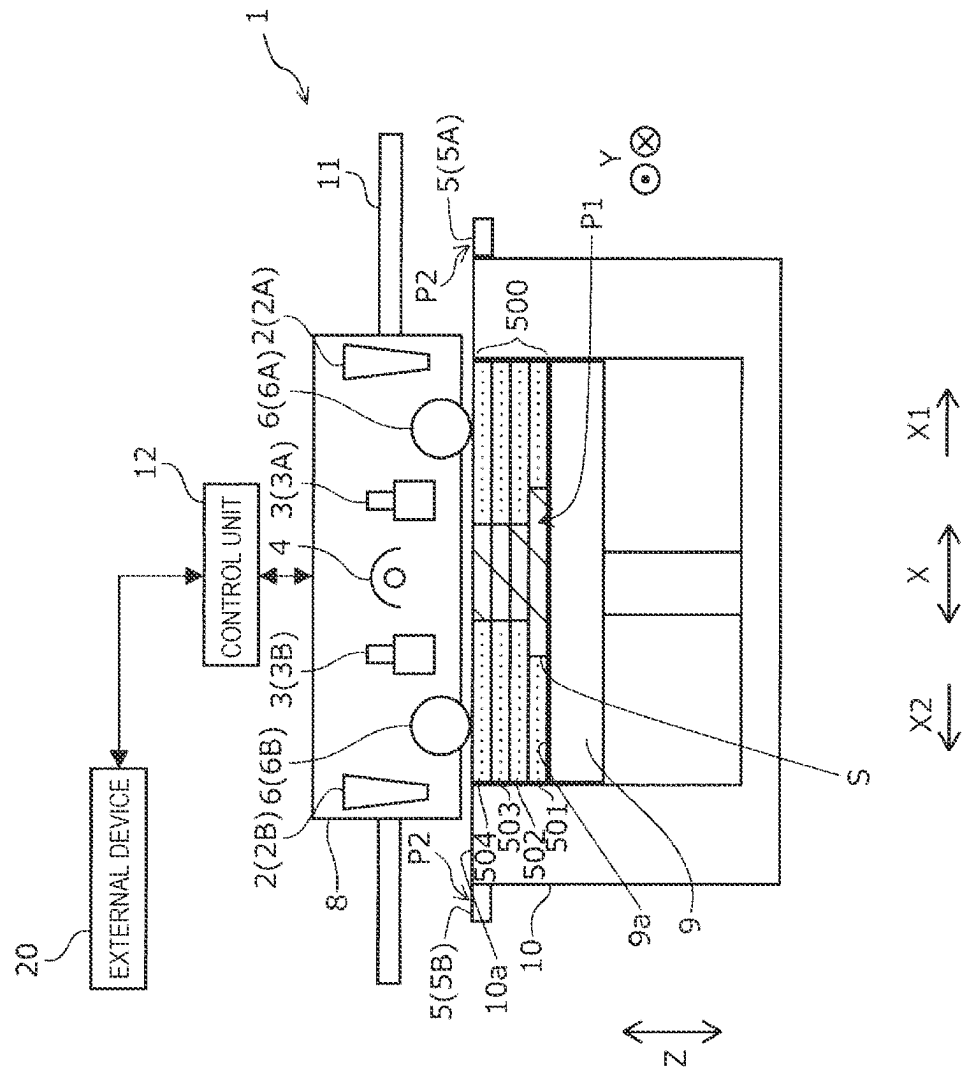
2015/0298394 A1 * 10/2015 Sheinman B29C 64/106
427/402
2016/0001504 A1 * 1/2016 Ikeda B29C 64/393
264/255
2017/0266963 A1 * 9/2017 Garcia Alvarez B41J 2/16526
2018/0170058 A1 * 6/2018 Shih B29C 64/00
2018/0178533 A1 * 6/2018 Sakai B22F 12/53
2018/0244065 A1 * 8/2018 Horade B41J 2/16517
2018/0354192 A1 12/2018 Iwase et al.
2019/0118258 A1 * 4/2019 Sachs B29C 64/393
2020/0061920 A1 * 2/2020 Debora B29C 64/393
2020/0086643 A1 * 3/2020 Imoto B41J 2/14088

FOREIGN PATENT DOCUMENTS

JP 2016060197 A * 4/2016 B29C 64/165
JP 2019-001010 1/2019
JP 2019077152 A * 5/2019 B29C 64/209
WO WO-2019097255 A1 * 5/2019 B22F 10/00

* cited by examiner

FIG. 1



2017

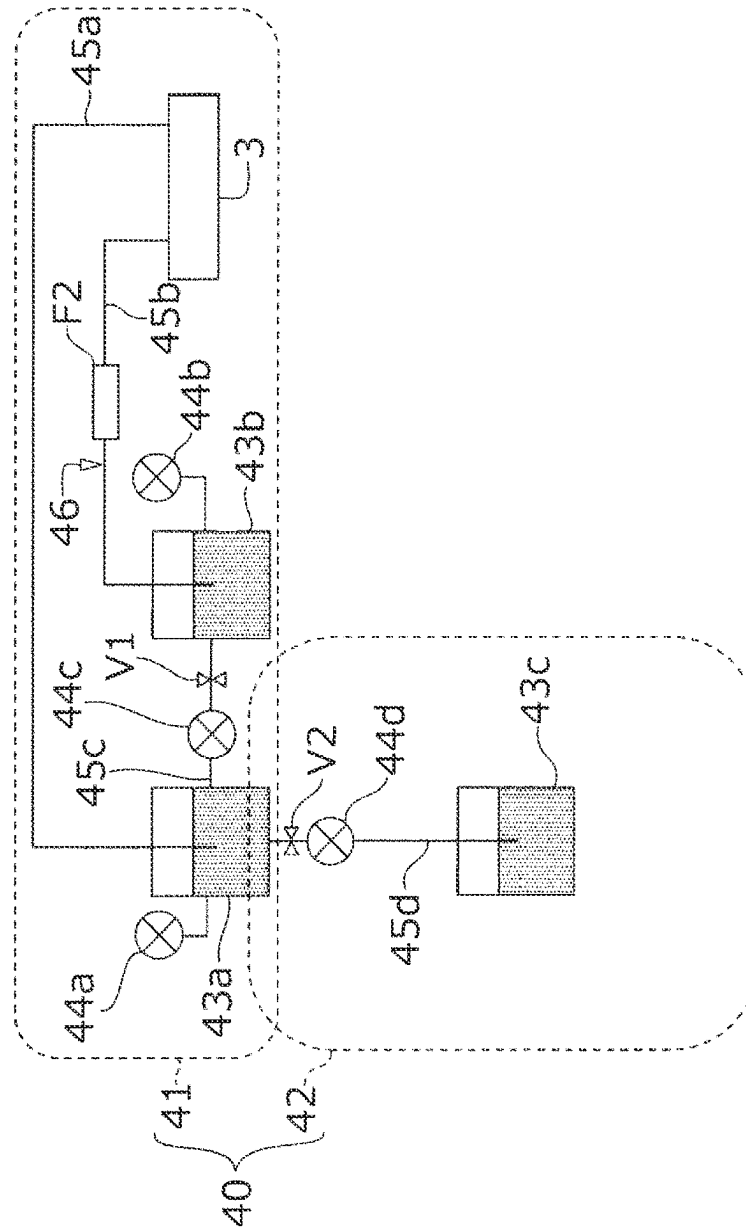


FIG. 3

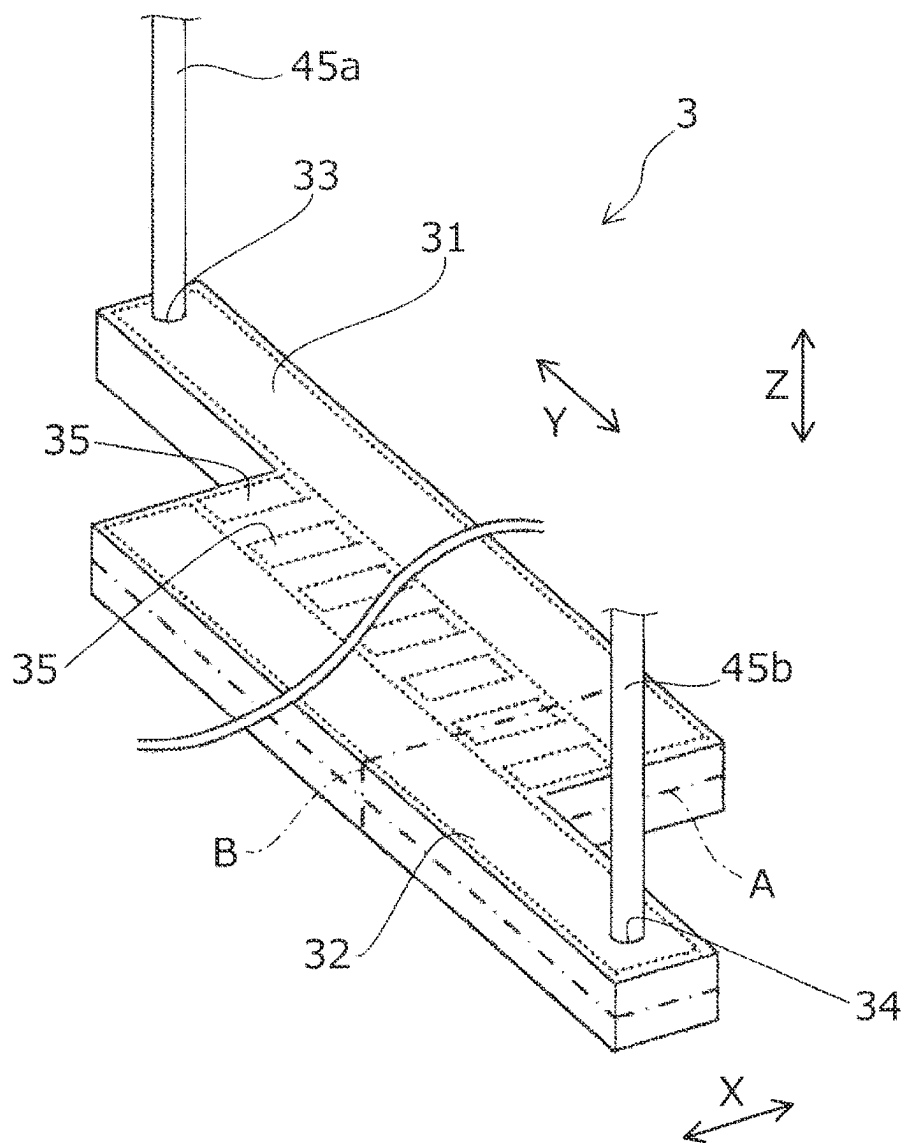


FIG. 4

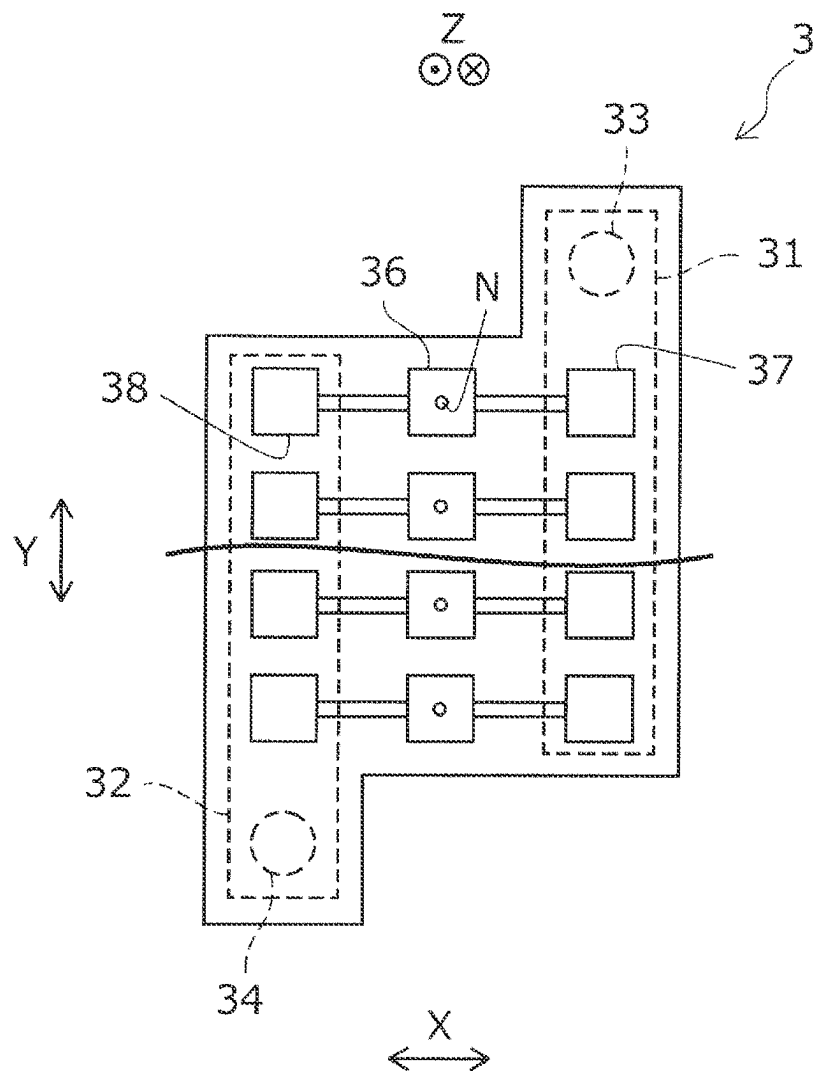


FIG. 5

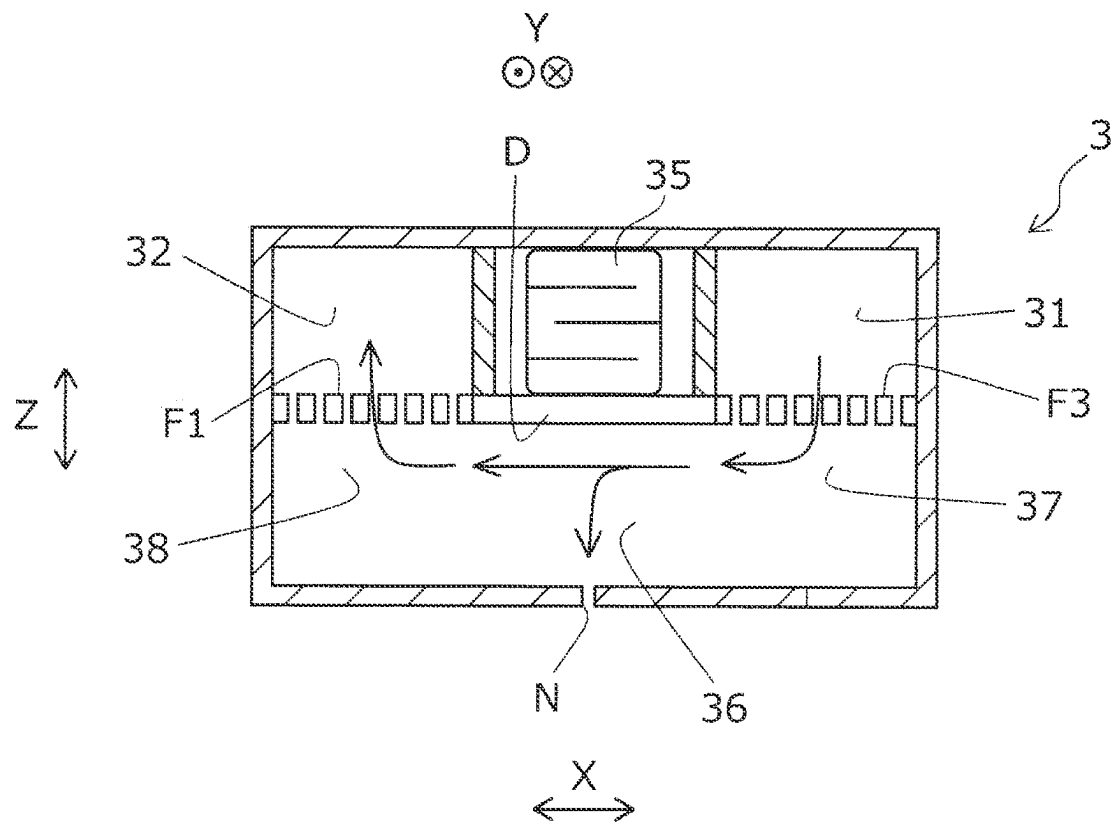


FIG. 6

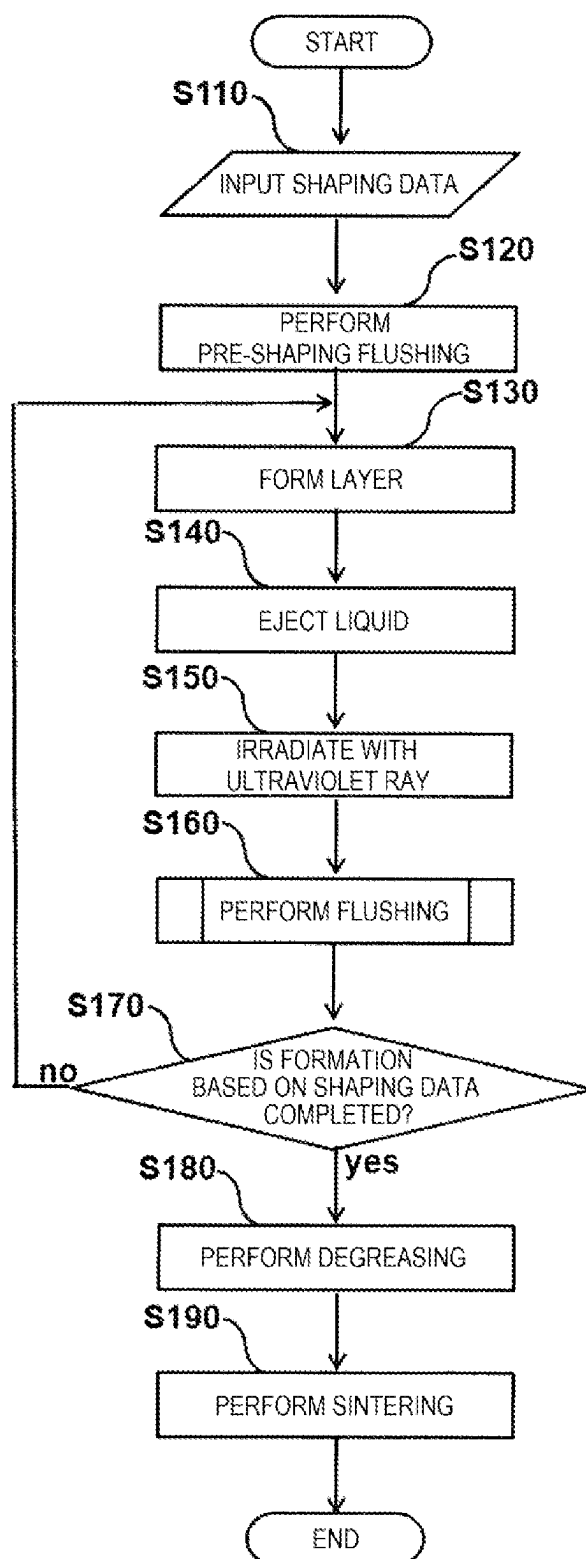


FIG. 7

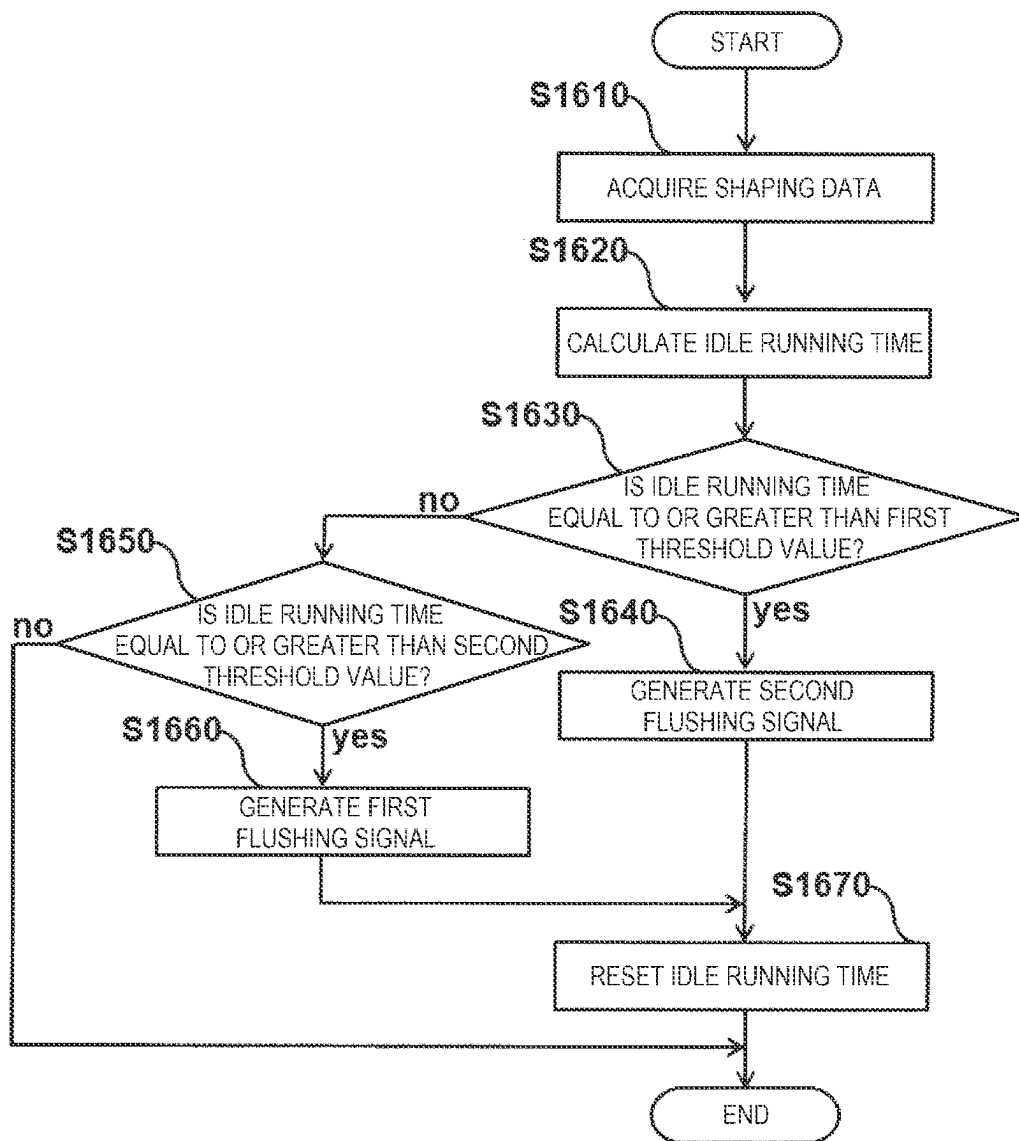


FIG. 8

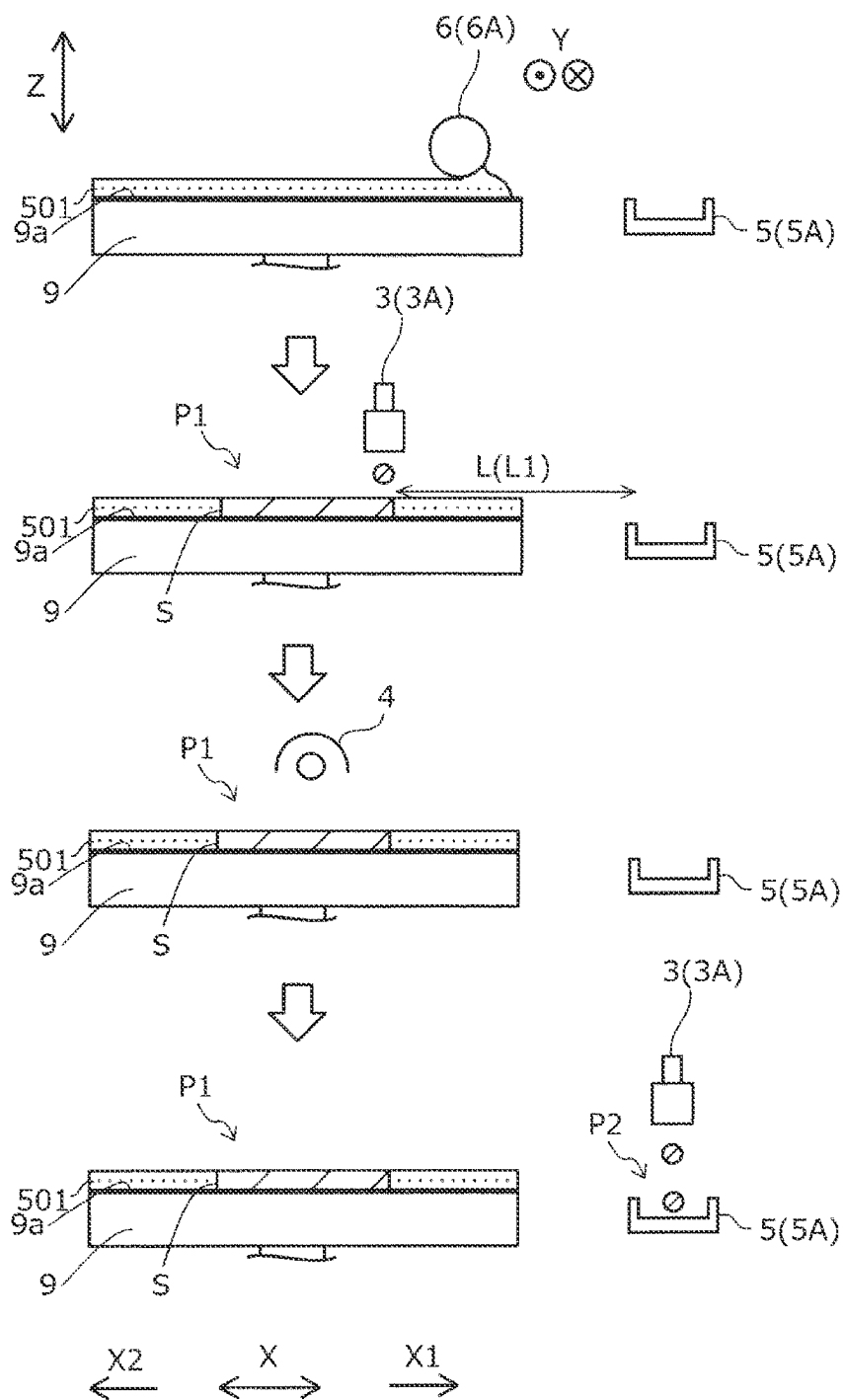


FIG. 9

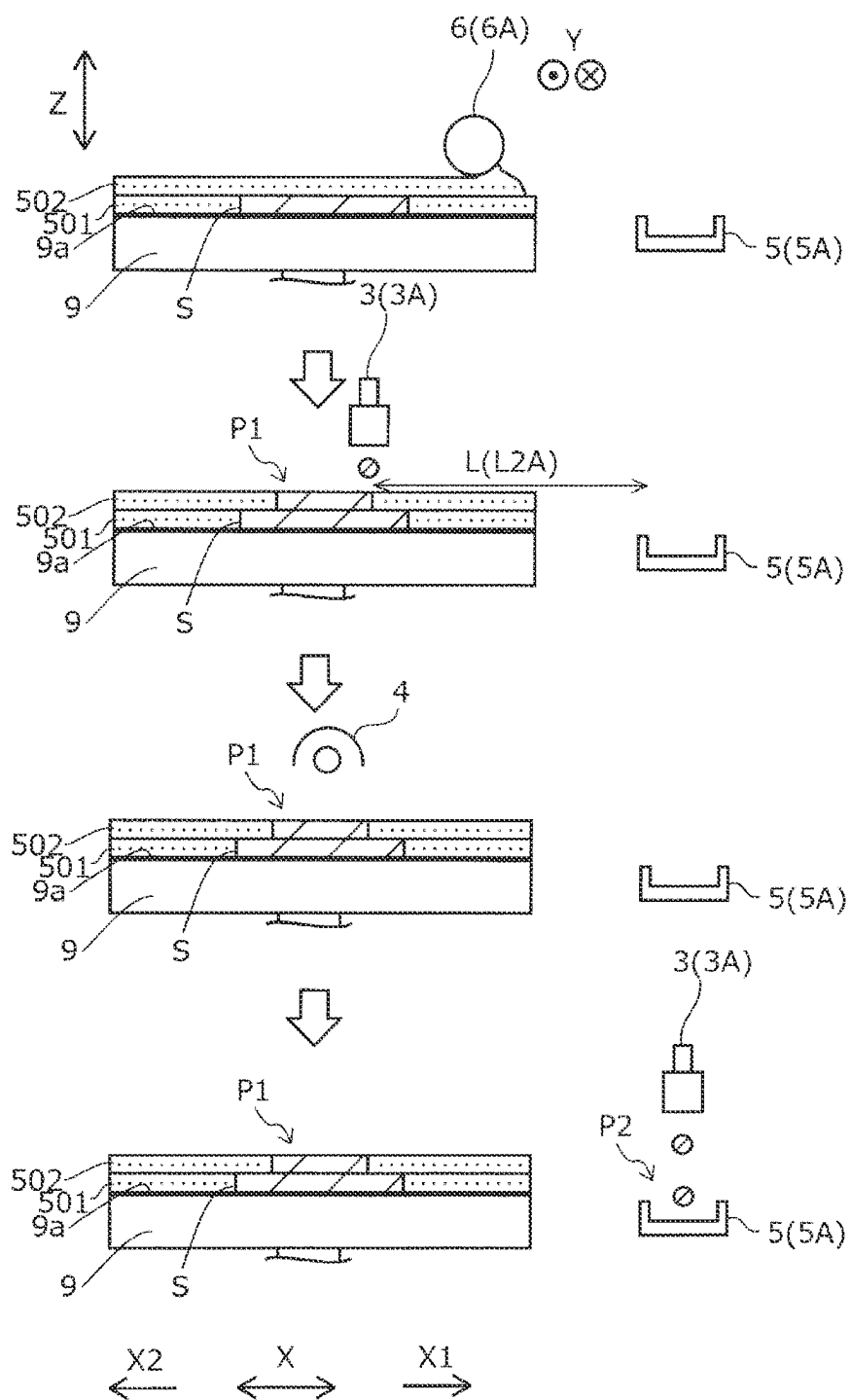


FIG. 10

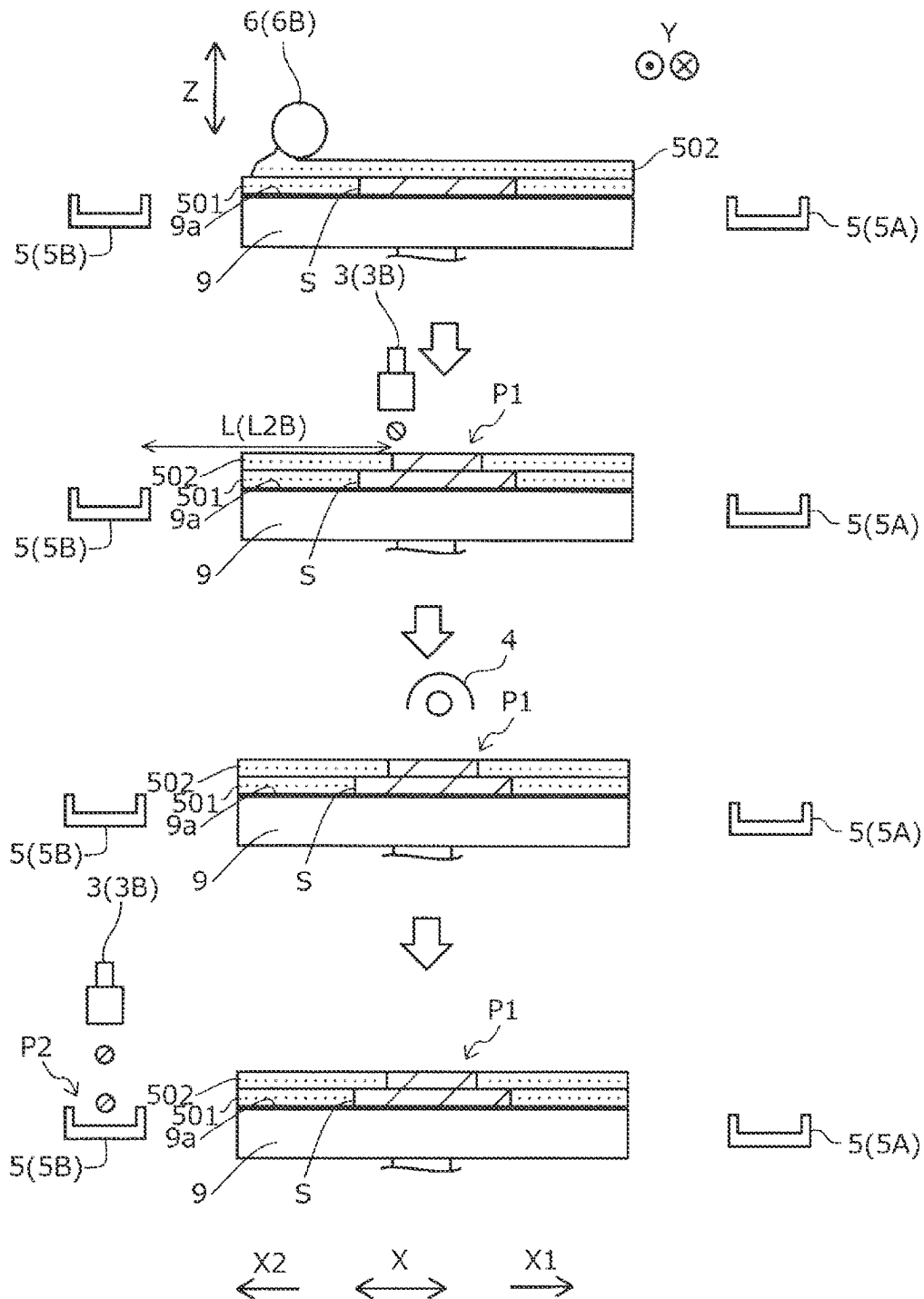


FIG. 11

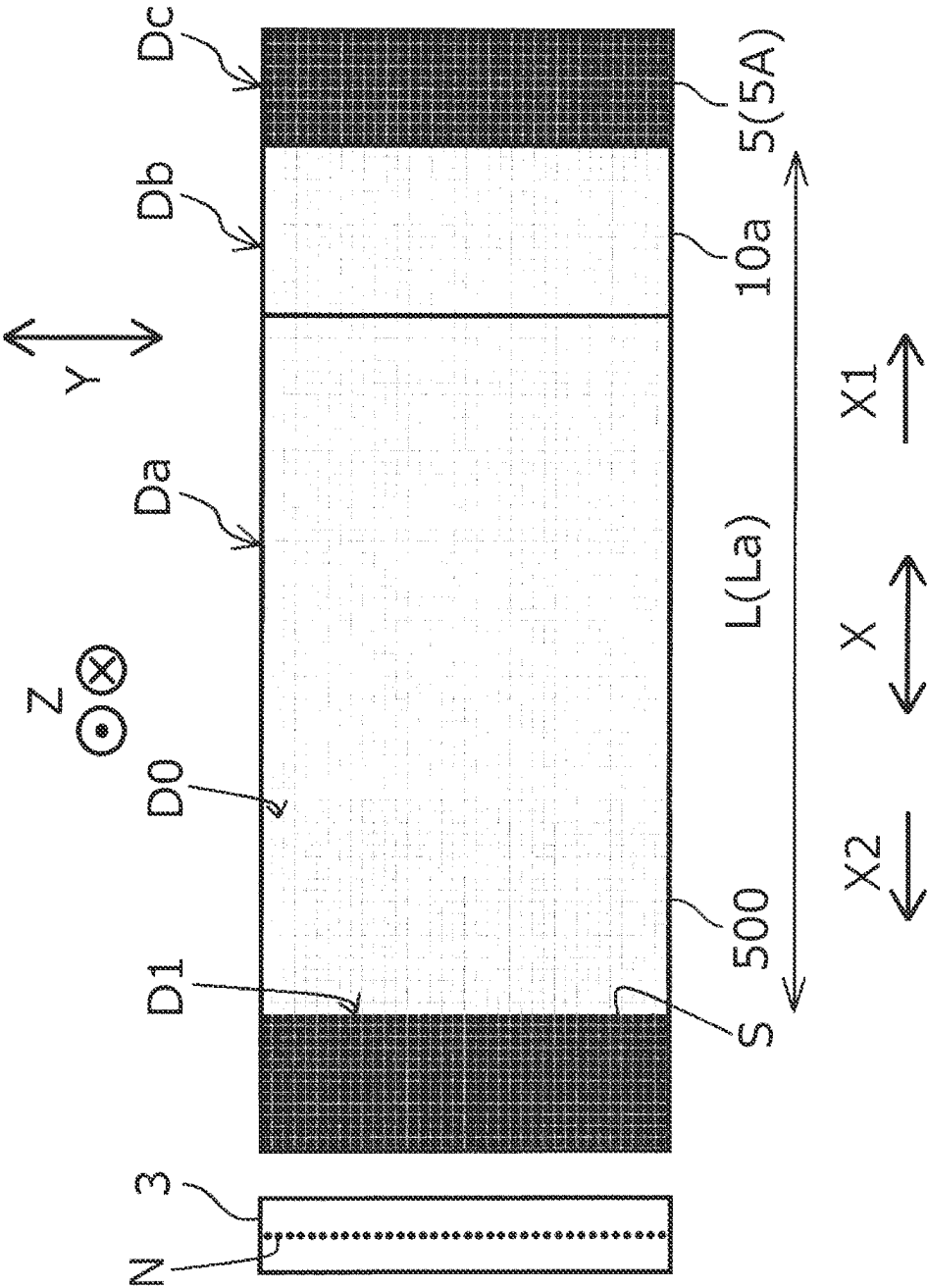


FIG. 12

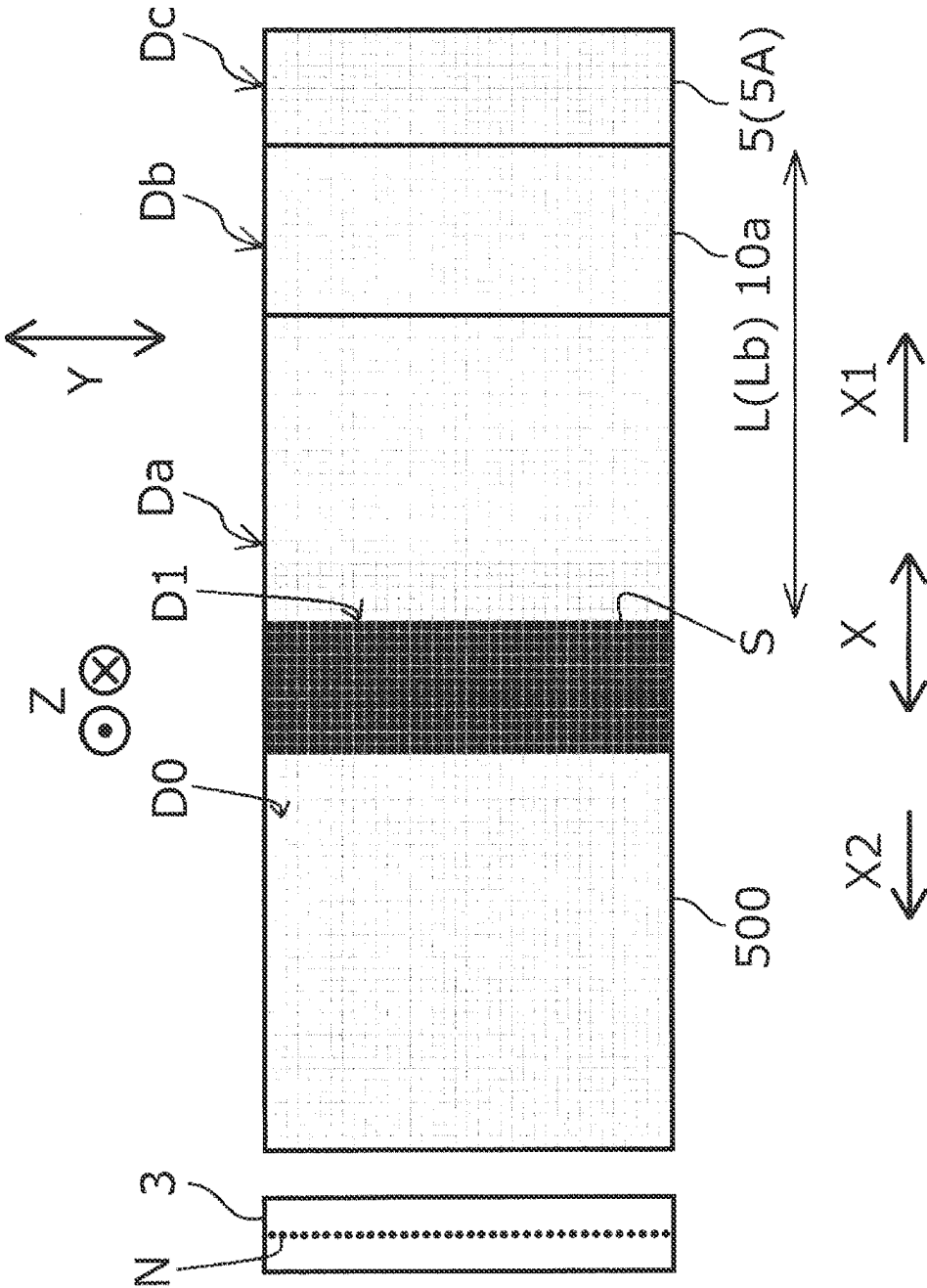


FIG. 13

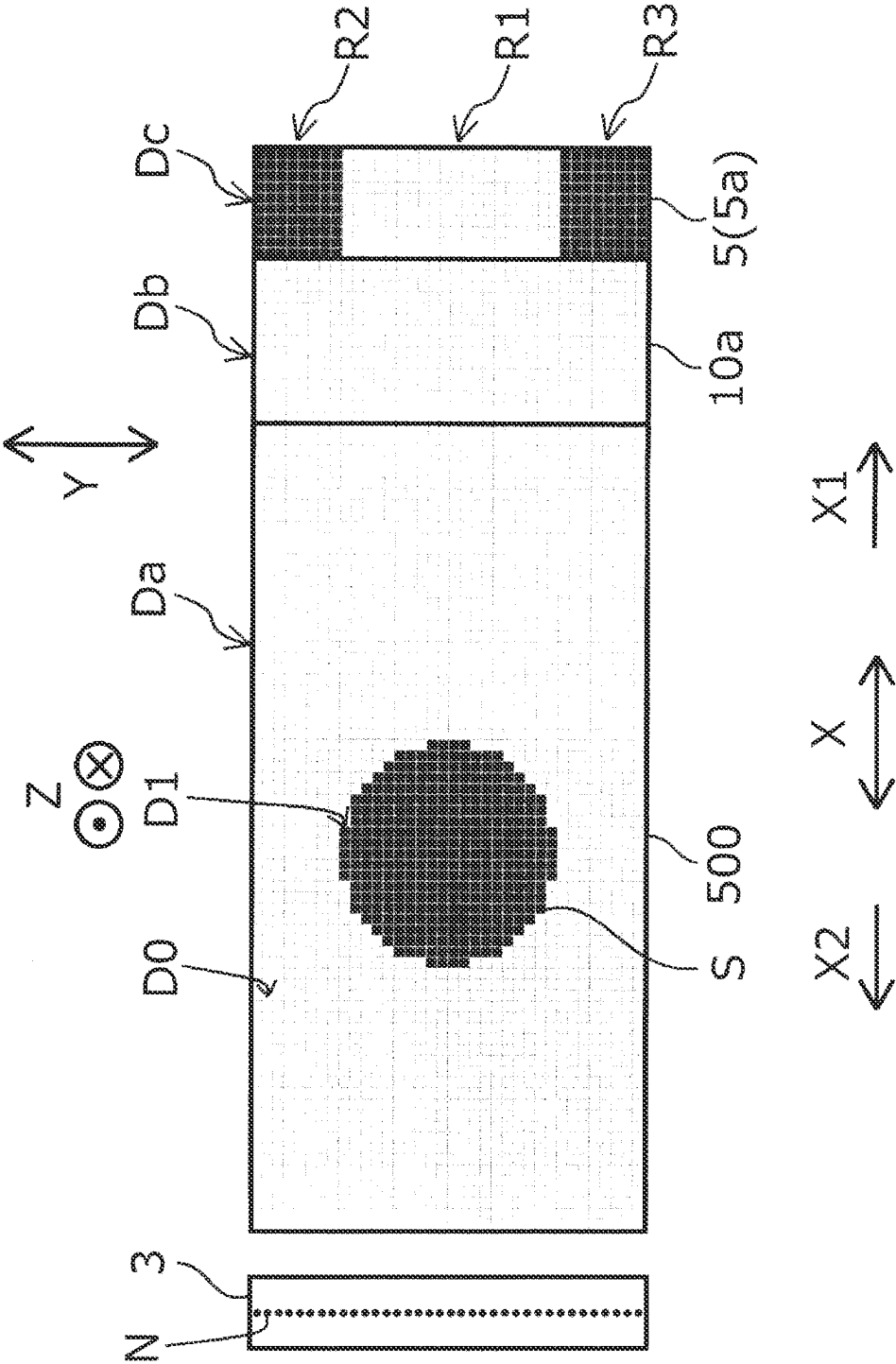


FIG. 14

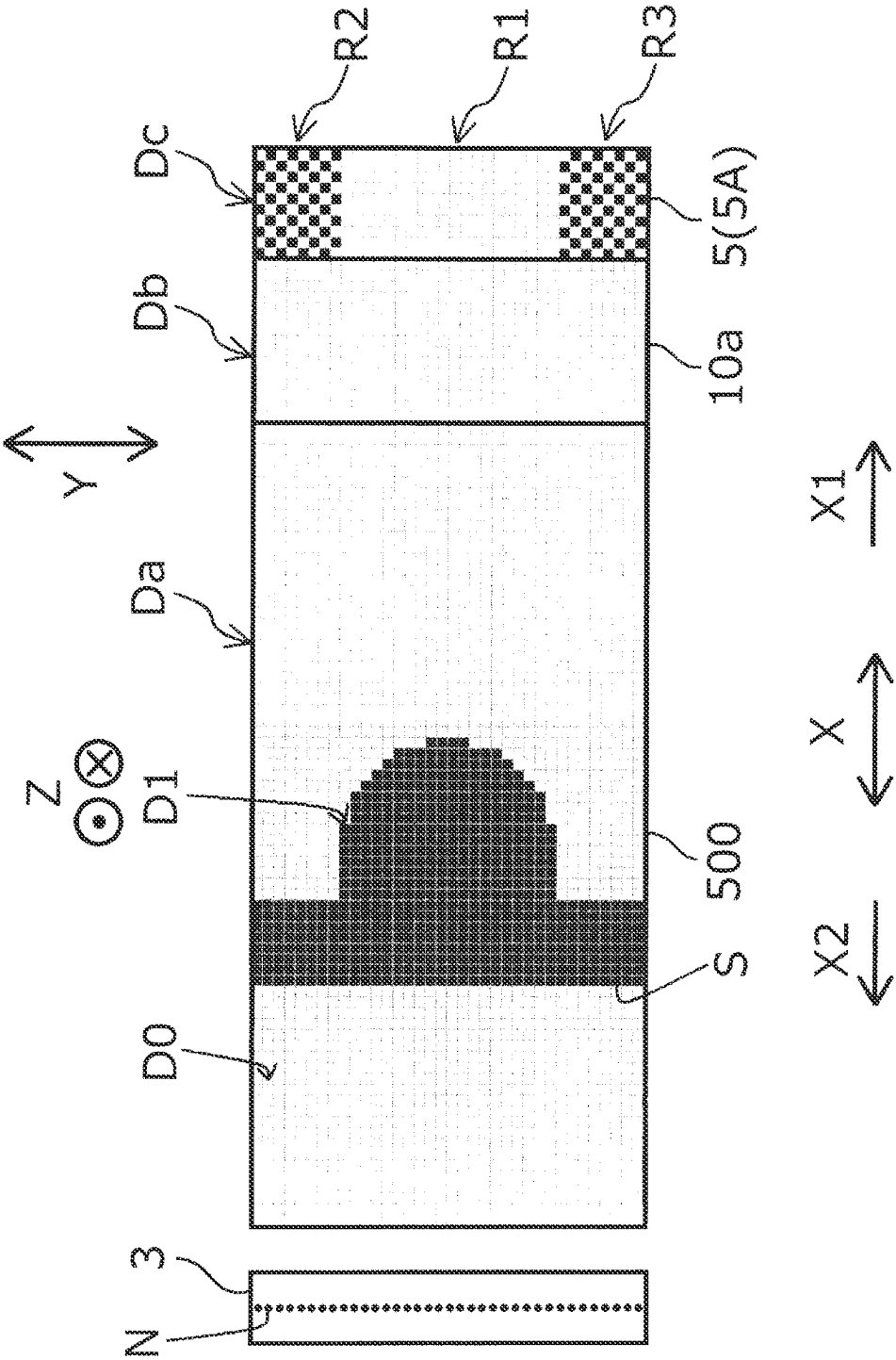
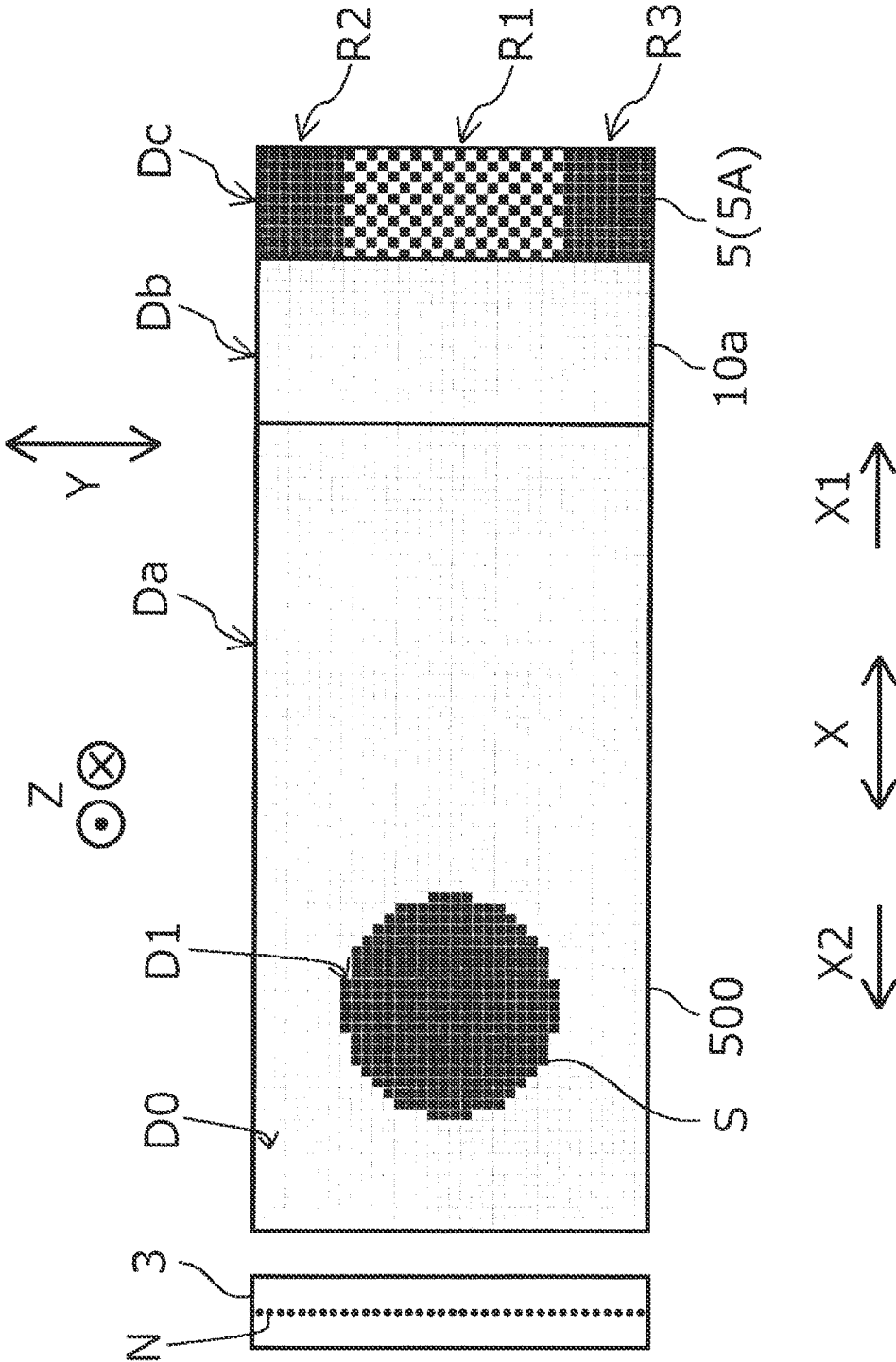


FIG. 15



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THREE-DIMENSIONAL SHAPING DEVICE WITH NOZZLE FLUSHING OPERATION

The present application is based on, and claims priority from JP Application Serial Number 2020-125895, filed Jul. 23, 2020, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to a three-dimensional shaping device.

2. Related Art

In related art, various types of three-dimensional shaping devices have been used. Among the devices, there is a three-dimensional shaping device that forms a powder layer and ejects a liquid containing a binder from a nozzle to a shaping region of a three-dimensional shaped object in the powder layer to manufacture a three-dimensional shaped object. For example, JP-A-2019-1010 discloses a three-dimensional shaping device that forms a layer of a powder material and ejects a curing liquid from a nozzle of a line head to the layer to manufacture a three-dimensional shaped object.

In the three-dimensional shaping device described in JP-A-2019-1010, a flushing stage for executing a flushing operation of ejecting a curing liquid from a nozzle is formed. A foreign substance in the nozzle is removed by the flushing operation. In order to simplify the control during the flushing operation, the flushing operation is generally performed under the same ejection conditions as when the three-dimensional shaped object is shaped, that is, when the liquid is ejected to a shaping region of the three-dimensional shaped object. However, in recent years, the three-dimensional shaped object is manufactured with various materials, and various kinds of powder are used. Therefore, depending on the powder to be used, a large amount of powder may float up and be mixed into the nozzle. The possibility of the powder floating up and mixed into the nozzle is increased as a time during which a head is moved on the powder layer in an idle running state without ejection becomes longer. When the powder mixed into the nozzle cannot be removed, a liquid ejection failure or the like may occur.

SUMMARY

In order to solve the above problems, a three-dimensional shaping device according to the present disclosure includes: a shaping table; a layer forming unit configured to form a powder layer on the shaping table; a head configured to eject a liquid containing a binder from a nozzle to a shaping region of a three-dimensional shaped object on the powder layer; and a control unit configured to control movement of the head with respect to the shaping table and driving of the head by applying a voltage, and to control the head to execute a flushing operation at a flushing position which is a position different from the shaping region, in which the control unit changes an execution condition of the flushing operation depending on whether an idle running time during which the head is moved without ejecting the liquid is less than a threshold value or equal to or greater than the threshold value.

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BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic configuration diagram showing a three-dimensional shaping device according to an embodiment of the present disclosure.

FIG. 2 is a schematic diagram showing a liquid sharing system of the three-dimensional shaping device in FIG. 1.

FIG. 3 is a perspective view showing a head of the three-dimensional shaping device in FIG. 1.

FIG. 4 is a cross-sectional view of the head of the three-dimensional shaping device in FIG. 1 taken along a dash-dotted line A in FIG. 3 as viewed from a bottom surface side, in which a part of constituent members are seen through and shown by broken lines.

FIG. 5 is a cross-sectional side view of the head of the three-dimensional shaping device in FIG. 1, taken along a dash-dotted line B in FIG. 3.

FIG. 6 is a flow chart of a three-dimensional shaping method performed by using the three-dimensional shaping device in FIG. 1.

FIG. 7 is a flow chart showing flushing processing in the flow chart in FIG. 6.

FIG. 8 is a schematic view illustrating the three-dimensional shaping method performed by using the three-dimensional shaping device in FIG. 1, and is a view showing a state where a first layer is formed.

FIG. 9 is a schematic view illustrating the three-dimensional shaping method performed by using the three-dimensional shaping device in FIG. 1, and is a view showing a state where a supply unit is moved in a direction the same as that in forming the first layer to form a second layer.

FIG. 10 is a schematic view illustrating the three-dimensional shaping method performed by using the three-dimensional shaping device in FIG. 1, and is a view showing a state where the supply unit is moved in a direction opposite to that in forming the first layer to form a second layer.

FIG. 11 is a diagram showing an example of a flushing position when a flushing operation is executed.

FIG. 12 is a diagram showing an example of a case where the flushing operation is not executed.

FIG. 13 is a diagram showing an example of a flushing position when the flushing operation is executed, which is different from the example shown in FIG. 11.

FIG. 14 is a diagram showing an example of a flushing position when the flushing operation is executed, which is different from the examples shown in FIGS. 11 and 13.

FIG. 15 is a diagram showing an example of a flushing position when the flushing operation is executed, which is different from the examples shown in FIGS. 11, 13 and 14.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

First, the present disclosure will be schematically described.

In order to solve the above problems, a three-dimensional shaping device according to a first aspect of the present disclosure includes: a shaping table; a layer forming unit configured to form a powder layer on the shaping table; a head configured to eject a liquid containing a binder from a nozzle to a shaping region of a three-dimensional shaped object on the powder layer; and a control unit configured to control movement of the head with respect to the shaping table and driving of the head by applying a voltage, and to control the head to execute a flushing operation at a flushing position which is a position different from the shaping region, in which the control unit changes an execution

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condition of the flushing operation depending on whether an idle running time during which the head is moved without ejecting the liquid is less than a threshold value or equal to or greater than the threshold value.

According to the present aspect, the execution condition of the flushing operation is changed depending on whether the idle running time is less than the threshold value or equal to or greater than the threshold value. For this reason, for example, when the idle running time becomes long and the time during which the head is moved on the powder layer in an idle running state becomes long, it is possible to discharge the liquid in the head under a condition where a flushing effect is high. Therefore, it is possible to prevent the powder mixed into the nozzle from not being removed even though the flushing operation is performed.

The three-dimensional shaping device according to a second aspect of the present disclosure is directed to the first aspect, in which the control unit is configured to generate flushing data related to ejection of the liquid at the flushing position based on shaping data of the three-dimensional shaped object, and control the head to execute a flushing operation based on the flushing data.

According to the present aspect, the flushing data is generated based on the shaping data, and the head is caused to execute the flushing operation based on the flushing data. The flushing operation can be appropriately executed by generating the flushing data from the shaping data.

The three-dimensional shaping device according to a third aspect of the present disclosure is directed to the first aspect or the second aspect, in which the control unit does not execute the flushing operation when the idle running time is less than a threshold value, and executes the flushing operation when the idle running time is equal to or greater than a threshold value.

According to the present aspect, the flushing operation is not executed when the idle running time is less than the threshold value. Therefore, it is possible to effectively prevent the liquid from being wasted due to the flushing operation being performed even when the flushing operation is unnecessary.

The three-dimensional shaping device according to a fourth aspect of the present disclosure is directed to the first aspect or the second aspect, in which the control unit controls the head to execute a first flushing operation when the idle running time is less than a threshold value, and controls the head to execute a second flushing operation under a flushing condition different from that of the first flushing operation when the idle running time is equal to or greater than the threshold value.

According to the present aspect, the first flushing operation is executed when the idle running time is less than the threshold value, and the second flushing operation is executed when the idle running time is equal to or greater than the threshold value. Therefore, for example, when the idle running time during which the head is moved on the powder layer in an idle running state becomes long, it is possible to discharge the liquid in the head under a condition where a flushing effect is high. Therefore, it is possible to prevent the powder mixed into the nozzle from not being removed even though the flushing operation is performed.

The three-dimensional shaping device according to a fifth aspect of the present disclosure is directed to the fourth aspect, in which a frequency of a waveform input to the head in the second flushing operation is higher than a frequency of a waveform input to the head in the first flushing operation.

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In general, when a distance between the shaping region and the flushing position becomes long and the time during which the head is moved on the powder layer becomes long, a larger amount of powder tends to be mixed into the nozzle, and it is difficult to discharge the powder mixed into the nozzle in many cases. However, according to the present aspect, the frequency of the waveform input to the head in the second flushing operation is higher than the frequency of the waveform input to the head in the first flushing operation. When the frequency of the waveform input to the head is increased, the powder mixed into the nozzle can be effectively discharged, and therefore, according to the present aspect, the powder mixed into the nozzle can be effectively discharged even when the idle running time becomes long.

The three-dimensional shaping device according to a sixth aspect of the present disclosure is directed to the fourth aspect or the fifth aspect, in which a voltage to be applied to the head in the second flushing operation is higher than a voltage to be applied to the head in the first flushing operation.

According to the present aspect, the voltage applied to the head in the second flushing operation is higher than the voltage applied to the head in the first flushing operation. When the voltage applied to the head is increased, the powder mixed into the nozzle can be effectively discharged, and therefore, according to the present aspect, the powder mixed into the nozzle can be effectively discharged even when the idle running time becomes long.

The three-dimensional shaping device according to a seventh aspect of the present disclosure is directed to any one of the fourth aspect to the sixth aspect, in which an ejection speed of the liquid from the head in the second flushing operation is faster than an ejection speed of the liquid from the head in the first flushing operation.

According to the present aspect, the ejection speed of the liquid from the head in the second flushing operation is faster than the ejection speed of the liquid from the head in the first flushing operation. When the ejection speed of the liquid from the head is increased, the powder mixed into the nozzle can be effectively discharged, and therefore, according to the present aspect, the powder mixed into the nozzle can be effectively discharged even when the idle running time becomes long.

The three-dimensional shaping device according to an eighth aspect of the present disclosure is directed to any one of the fourth aspect to the seventh aspect, in which a droplet size of the liquid ejected from the head in the second flushing operation is larger than a droplet size of the liquid ejected from the head in the first flushing operation.

According to the present aspect, the droplet size of the liquid ejected from the head in the second flushing operation is larger than the droplet size of the liquid ejected from the head in the first flushing operation. When an ejection amount of the liquid from the head is increased, the powder mixed into the nozzle can be effectively discharged, and therefore, according to the present aspect, the powder mixed into the nozzle can be effectively discharged even when the idle running time becomes long.

The three-dimensional shaping device according to a ninth aspect of the present disclosure is directed to any one of the fourth aspect to the eighth aspect, in which the control unit controls to execute the first flushing operation at least when operation of the three-dimensional shaping device is started or when the operation of the three-dimensional shaping device is completed.

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According to the present aspect, the first flushing operation is executed at least when the operation of the three-dimensional shaping device is started or when the operation of the three-dimensional shaping device is completed. Therefore, when the three-dimensional shaping device is operated, it is possible to prevent occurrence of a liquid ejection failure due to the powder mixed into the nozzle.

The three-dimensional shaping device according to a tenth aspect of the present disclosure is directed to any one of the first aspect to the ninth aspect, in which the control unit resets a measurement start timing of the idle running time every time one powder layer is formed.

According to the present aspect, the measurement start timing of the idle running time is reset every time one powder layer is formed. Therefore, the powder mixed into the nozzle can be effectively discharged by simple control.

The three-dimensional shaping device according to an eleventh aspect of the present disclosure is directed to any one of the first aspect to the ninth aspect, in which the control unit resets a measurement start timing of the idle running time every time the flushing operation is executed.

According to the present aspect, the measurement start timing of the idle running time is reset every time the flushing operation is executed. Therefore, since the number of times of the flushing operation can be reduced, it is possible to effectively discharge the powder mixed into the nozzle while effectively preventing waste of the liquid.

The three-dimensional shaping device according to a twelfth aspect of the present disclosure is directed to any one of the first aspect to the eleventh aspect, in which the head includes a pressure chamber communicating with the nozzle, a supply path configured to supply the liquid to the pressure chamber, and a circulation path into which the liquid from the pressure chamber flows for circulation.

According to the present aspect, the head includes the circulation path for circulating the liquid. Therefore, by circulating the liquid, it is possible to prevent precipitation of a solid component contained in the liquid, and it is possible to prevent a problem associated with precipitation of the solid component.

The three-dimensional shaping device according to a thirteenth aspect of the present disclosure is directed to the twelfth aspect, in which the control unit is configured to control such that q_2/q_1 is equal to or greater than 0.05 and equal to or less than 20, where q_1 is a flow rate of the liquid flowing into the circulation path per unit time, and q_2 is a maximum flow rate of the liquid ejected from the nozzle per unit time.

According to the present aspect, the flow rate of the liquid flowing into the circulation path is controlled within an appropriate range. As a result, it is possible to prevent an increase in a pressure difference between a pressure applied to the liquid in the nozzle and an outside air pressure. By preventing an increase in the pressure difference, it is possible to prevent mixing of powder into the nozzle.

The three-dimensional shaping device according to a fourteenth aspect of the present disclosure is directed to the thirteenth aspect, in which q_2/q_1 is 0.05.

According to the present aspect, the flow rate of the liquid flowing into the circulation path is controlled within a particularly preferable range. As a result, it is possible to particularly effectively prevent an increase in the pressure difference between the pressure applied to the liquid in the nozzle and the outside air pressure, and it is possible to particularly effectively prevent mixing of the powder into the nozzle.

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Hereinafter, an embodiment of the present disclosure will be described with reference to accompanying drawings.

First, an embodiment of a three-dimensional shaping device 1 according to the present disclosure will be described with reference to FIG. 1. Here, in FIG. 1 and the drawings to be described later, an X direction is a horizontal direction and corresponds to a reciprocating direction of a supply unit 8, and an X1 direction corresponds to a forward direction and an X2 direction corresponds to a backward direction. A Y direction is a horizontal direction and a direction orthogonal to the X direction, and corresponds to a direction in which a rotation shaft of a roller 6 extends. A Z direction is a vertical direction and corresponds to a lamination direction of layers 500.

The term “three-dimensional shaping” in this description refers to forming a so-called three-dimensional shaped object, and includes, for example, forming a shape having a thickness even in a shape of a flat plate shape that is a so-called two-dimensional shape.

The three-dimensional shaping device 1 according to the present embodiment is a three-dimensional shaping device that manufactures a three-dimensional shaped object by laminating the layers 500 including a layer 501, a layer 502, a layer 503, . . . , and a layer 50n. As shown in FIG. 1, the three-dimensional shaping device 1 according to the present embodiment includes a table unit 10 that includes a shaping table 9, the supply unit 8 that supplies a shaping material of the three-dimensional shaped object to the shaping table 9, and a control unit 12 that controls operations of the table unit 10 and the supply unit 8. The three-dimensional shaping device 1 is electrically coupled to an external device 20 such as a personal computer, and is configured to receive an instruction from a user via the external device 20.

The shaping table 9 is configured to be movable in the Z direction under control of the control unit 12. A shaping surface 9a of the shaping table 9 is disposed at a position lower than an upper surface portion 10a of the table unit 10 by a predetermined distance in the Z direction, and the shaping material of the three-dimensional shaped object is supplied from the supply unit 8 to the shaping surface 9a to form one layer of the layers 500. Then, laminating is performed by repeating a downward movement of the shaping table 9 by a predetermined distance and a supply of the shaping material of the three-dimensional shaped object from the supply unit 8. FIG. 1 shows a state where a structure S of the three-dimensional shaped object is formed at the shaping surface 9a by repeating a formation of four layers of the layer 501, the layer 502, the layer 503, and a layer 504.

The supply unit 8 is configured to be movable in the X direction along a guide bar 11. In addition, the supply unit 8 includes shaping material supply units 2 that supply the shaping material including powder of metal, ceramics, resin, or the like to the shaping table 9. The shaping material supply units 2 include a shaping material supply unit 2A formed at a head side end portion in the X1 direction and a shaping material supply unit 2B formed at a head side end portion in the X2 direction.

The supply unit 8 includes the rollers 6 that can compress and level the shaping material supplied to the shaping table 9. The rollers 6 include a roller 6A formed next to the shaping material supply unit 2A in the X direction and a roller 6B formed next to the shaping material supply unit 2B in the X direction. Here, the shaping material supply units 2 and the rollers 6 constitute a layer forming unit that forms the layers 500, which are powder layers, at the shaping table

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9. The supply unit 8 may include a squeegee that can level the shaping material supplied to the shaping table 9 instead of the roller 6.

The supply unit 8 includes heads 3 that eject, to a shaping region P1 of the three-dimensional shaped object, a liquid containing a binder for binding powder contained in the shaping material supplied from the shaping material supply units 2. The heads 3 include a head 3A formed next to the roller 6A in the X direction and a head 3B formed next to the roller 6B in the X direction. Here, the liquids ejected from the head 3A and the head 3B are the same liquid, and both are liquids containing an ultraviolet curing resin as a binder. It is noted that the liquid is not limited to such a liquid, and a liquid containing a thermosetting resin as a binder, a liquid in which a solid resin as a binder is dissolved in a volatile solvent, or the like may be used.

An ultraviolet ray irradiation unit 4 that performs irradiation with ultraviolet rays that can cure the ultraviolet curing resin is provided between the head 3A and the head 3B in the X direction. The supply unit 8 according to the present embodiment includes one ultraviolet ray irradiation unit 4, but may include two or more ultraviolet ray irradiation units 4, or not include the ultraviolet ray irradiation unit 4 according to a type of the liquid to be used, or include a heater for curing the thermosetting resin or volatilizing the solvent instead of the ultraviolet ray irradiation unit 4, or the like.

As shown in FIG. 1, in the supply unit 8 according to the present embodiment, shapes of constituent members are symmetrical in the X direction. Therefore, the three-dimensional shaping device 1 according to the present embodiment can execute a shaping operation of the three-dimensional shaped object while moving the supply unit 8 in the X1 direction, and execute the shaping operation of the three-dimensional shaped object while moving the supply unit 8 in the X2 direction.

Further, as shown in FIG. 1, in the three-dimensional shaping device 1 according to the present embodiment, liquid receiving units 5 are provided on the table unit 10, and a flushing operation is executed by ejecting a liquid from the head 3 at a position facing the liquid receiving unit 5. That is, the position facing the liquid receiving unit 5 is a flushing position P2, and thus the flushing position P2 is certainly different from the shaping region P1 of the three-dimensional shaped object. The liquid receiving units 5 include a liquid receiving unit 5A and a liquid receiving unit 5B. A detailed flushing area on the liquid receiving unit 5 at the flushing position P2 will be described later.

As described above, the three-dimensional shaping device 1 according to the present embodiment includes the shaping table 9, the shaping material supply units 2 and the rollers 6 which are serving as the layer forming unit that forms, at the shaping table 9, the layers 500 which are powder layers, the heads 3 that eject, from a nozzle N, a liquid containing a binder to the shaping region P1 of the three-dimensional shaped object in the layers 500, and the control unit 12 that controls movements of the heads 3 with respect to the shaping table 9 and driving of the heads 3 by applying a voltage. After the liquid is ejected to the shaping region P1, the control unit 12 applies a voltage to the head 3 at the flushing position that is different from the shaping region P1 to execute the flushing operation of ejecting the liquid from the nozzle N. In the three-dimensional shaping device 1 according to the present embodiment, the position facing the liquid receiving unit 5 is defined as the flushing position P2, but the flushing position is not limited thereto, and, for

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example, a region different from the shaping region P1 on the shaping surface 9a may be used as the flushing position P2.

The three-dimensional shaping device 1 according to the present embodiment includes a liquid supply system. 40 that supplies the liquid to the head 3. Hereinafter, the liquid supply system 40 and the head 3 will be described in detail with reference to FIGS. 2 to 5. Here, the liquid supply system 40 shown in FIG. 2 includes a circulation unit 41 including a supply flow path 45a for supplying the liquid to the head 3, and a replenishment unit 42 including a liquid replenishment flow path 45d for replenishing the circulation unit 41 with a liquid.

First, the liquid supply system 40 will be described. As shown in FIG. 2, the circulation unit 41 includes the heads 3, a pressure increase control liquid tank 43a, a pressure reduction control liquid tank 43b, a pressure increase control pump 44a, a pressure reduction control pump 44b, a flow pump 44c, and an electromagnetic valve V1. The circulation unit 41 includes a supply flow path 45a that couples the pressure increase control liquid tank 43a and the head 3, a first circulation flow path 45b that couples the head 3 and the pressure reduction control liquid tank 43b, and a second circulation flow path 45c that couples the pressure increase control liquid tank 43a and the pressure reduction control liquid tank 43b. Here, the first circulation flow path 45b is provided with a filter F2 and a flow rate sensor 46 that detects a flow rate of the liquid flowing through the first circulation flow path 45b.

Differential pressure control is performed by the pressure increase control liquid tank 43a, the pressure increase control pump 44a, the pressure reduction control liquid tank 43b, and the pressure reduction control pump 44b so that a slight negative pressure from an atmospheric pressure is applied to the nozzle N of the head 3.

The flow pump 44c and the electromagnetic valve V1 are installed in the second circulation flow path 45c for causing the liquid to flow from the pressure reduction control liquid tank 43b, which is a pressure reduction tank, to the pressure increase control liquid tank 43a, which is a pressure increase tank. When a liquid ejection operation in the head 3 is executed and the liquid is supplied to the head 3, the electromagnetic valve V1 is opened and the flow pump 44c is operated to circulate the liquid in the supply flow path 45a, the first circulation flow path 45b, and the second circulation flow path 45c.

The replenishment unit 42 includes a replaceable liquid cartridge 43c in which a liquid is stored, a flow pump 44d, and an electromagnetic valve V2. In addition, the replenishment unit 42 includes the liquid replenishment flow path 45d that couples the pressure increase control liquid tank 43a and the liquid cartridge 43c. When the pressure increase control liquid tank 43a is replenished with the liquid from the liquid cartridge 43c, the electromagnetic valve V2 is opened, and the flow pump 44d is operated to cause the liquid to flow in the liquid replenishment flow path 45d.

Next, a detailed configuration of the head 3 will be described with reference to FIGS. 3 to 5. Solid arrows in FIG. 5 indicate a direction in which the liquid flows inside the head 3.

As shown in FIG. 3, the head 3 is coupled to the supply flow path 45a and the first circulation flow path 45b. The supply flow path 45a as a supply flow path for supplying the liquid to the inside of the head 3 and the first circulation flow path 45b as a circulation flow path for temporarily discharging the liquid inside the head 3 to the outside and circulating the liquid can be regarded as constituting a part of the head

3. In other words, the head 3 includes the supply flow path 45a and the first circulation flow path 45b. The supply flow path 45a is coupled to a supply port 33, and the first circulation flow path 45b is coupled to a discharge port 34.

As shown in FIGS. 3 to 5, the head 3 includes a supply liquid chamber 31 including the supply port 33, and the liquid is sent from the supply flow path 45a to the supply liquid chamber 31 via the supply port 33. As shown in FIGS. 4 and 5, the head 3 includes an individual supply flow path 37 communicating with the supply liquid chamber 31 via a filter F3, and the liquid supplied to the supply liquid chamber 31 is sent to the individual supply flow path 37.

As shown in FIGS. 3 and 5, the head 3 includes piezoelectric elements 35 that is deformed along the Z direction when a voltage is applied, and the piezoelectric elements 35 are disposed in a space opposite to the pressure chamber 36 with a diaphragm D interposed therebetween in the Z direction. As shown in FIGS. 4 and 5, the pressure chamber 36 communicates with the individual supply flow path 37, and the liquid is sent from the individual supply flow path 37 to the pressure chamber 36. Further, the nozzle N communicates with the pressure chamber 36, a volume of the pressure chamber 36 is contracted by deformation of the piezoelectric elements 35, and the liquid in the pressure chamber 36 is pressurized, whereby the liquid is ejected from the nozzle N. A downward direction in FIG. 5 is a vertical downward direction, and an ejection direction of the liquid from the nozzle N is the vertical downward direction corresponding to a gravity direction.

As described above, the three-dimensional shaping device 1 according to the present embodiment includes the liquid supply system 40 shown in FIG. 2, and circulates and supplies the liquid to be supplied to the head 3. Therefore, in order to circulate the liquid once sent to the pressure chamber 36, the pressure chamber 36 also communicates with an individual circulation flow path 38 in addition to the individual supply flow path 37. The individual circulation flow path 38 communicates with a circulation liquid chamber 32 including the discharge port 34 via a filter F1. The three-dimensional shaping device 1 according to the present embodiment circulates the liquid by causing the liquid to flow through the supply flow path 45a, the supply liquid chamber 31, the individual supply flow path 37, the pressure chamber 36, the individual circulation flow path 38, the circulation liquid chamber 32, and the first circulation flow path 45b, inside the head 3.

As described above, the head 3 includes the pressure chamber 36 communicating with the nozzle N; the supply flow path 45a, the supply liquid chamber 31, and the individual supply flow path 37 which are serving as a supply path for supplying the liquid containing the binder to the pressure chamber 36; and the individual circulation flow path 38, the circulation liquid chamber 32, and the first circulation flow path 45b which are serving as a circulation path into which the liquid containing the binder flows from the pressure chamber 36 for circulation. Here, the liquid containing the binder may contain a solid component or the like, and as described above, the head 3 is provided with the circulation path for circulating the liquid containing the binder. Therefore, by circulating the liquid, it is possible to prevent precipitation of the solid component contained in the liquid, and it is possible to prevent a problem associated with the precipitation of the solid component.

The control unit 12 controls such that q_2/q_1 is equal to or greater than 0.05 and equal to or less than 20, where q_1 is a flow rate of the liquid containing the binder and flowing into the circulation path per unit time and q_2 is a maximum

flow rate of the liquid containing the binder and ejected from the nozzle N per unit time. That is, the three-dimensional shaping device 1 according to the present embodiment controls the flow rate of the liquid containing the binder and flowing into the circulation path within an appropriate range. As a result, it is possible to prevent an increase in a pressure difference between a pressure applied to the liquid in the nozzle N and an outside air pressure. The three-dimensional shaping device 1 according to the present embodiment can prevent mixing of the powder forming the layers 500 into the nozzle N by preventing an increase in the pressure difference.

In a normal state, the control unit 12 sets q_2/q_1 to 0.05. That is, the three-dimensional shaping device 1 according to the present embodiment controls the flow rate of the liquid containing the binder and flowing into the circulation path within a particularly preferable range. As a result, the three-dimensional shaping device 1 according to the present embodiment can particularly effectively prevent an increase in the pressure difference between the pressure applied to the liquid in the nozzle N and the outside air pressure, and can particularly effectively prevent mixing of the powder forming the layers 500 into the nozzle N.

Next, a specific example of the shaping material that can be used by the three-dimensional shaping device 1 according to the present embodiment will be described. As a metal powder that can be contained in the shaping material forming the layers 500, for example, a single powder of magnesium (Mg), iron (Fe), cobalt (Co), chromium (Cr), aluminum (Al), titanium (Ti), copper (Cu), or nickel (Ni), a powder of an alloy containing one or more of these metals (maraging steel, stainless steel (SUS), cobalt chromium molybdenum, a titanium alloy, a nickel alloy, an aluminum alloy, a cobalt alloy, or a cobalt chromium alloy), or a mixed powder thereof can be used.

As a ceramic powder that can be contained in the shaping material forming the layers 500, for example, silicon dioxide (SiO_2), titanium dioxide (TiO_2), aluminum oxide (Al_2O_3), zirconium oxide (ZrO_2), silicon nitride (Si_3N_4) can be preferably used.

As a resin particle that can be contained in the shaping material forming the layers 500 or the binder contained in the liquid to be ejected from the head 3 to the shaping region P1, for example, PMMA (acrylic), ABS (acrylonitrile-butadiene-acrylic acid ester), ASA (acrylonitrile-styrene-acrylic acid ester), PLA (polylactic acid), PEI (polyetherimide), PC (polycarbonate), PP (polypropylene), PE (polyethylene), PA (polyamide), EP (epoxy), PPS (polyphenylene sulfide), PS (polystyrene), paraffin wax, PVA (polyvinyl alcohol), carboxymethyl cellulose, polyoxymethylene, polymethyl methacrylate can be preferably used. In addition, for example, an acrylic resin, an epoxy resin, a silicone resin, a cellulose-based resin, or other synthetic resins can be used alone or in combination. Further, a thermoplastic resin, an ultraviolet curing resin of a type using radical polymerization of an unsaturated double bond such as acrylic, or a type using cationic polymerization such as epoxy can also be used.

Examples of a solvent contained in the liquid ejected from the head 3 include water; (poly) alkylene glycol monoalkyl ethers such as ethylene glycol monomethyl ether, ethylene glycol monoethyl ether, propylene glycol monomethyl ether, and propylene glycol monoethyl ether; acetic acid esters such as ethyl acetate, n-propyl acetate, iso-propyl acetate, n-butyl acetate, and iso-butyl acetate; aromatic hydrocarbons such as benzene, toluene, and xylene; ketones such as methyl ethyl ketone, acetone, methyl isobutyl ketone, ethyl-n-butyl ketone, diisopropyl ketone, and acetylacetone; alco-

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hols such as ethanol, propanol, and butanol; tetraalkylammonium acetates; sulfoxide-based solvents such as dimethyl sulfoxide and diethyl sulfoxide; pyridine-based solvents such as pyridine, γ -picoline, and 2,6-lutidine; and ionic liquids such as tetraalkylammonium acetate (for example, tetrabutylammonium acetate), and one or two or more selected from these can be used in combination.

Next, an example of a three-dimensional shaping method that can be executed by using the three-dimensional shaping device 1 according to the present embodiment will be described with reference to FIGS. 8 to 15 and using flow charts in FIGS. 6 and 7. The flow chart in FIG. 7 specifically shows the flushing processing in step S160 in the flow chart in FIG. 6. The three-dimensional shaping method according to the present embodiment shown by the flow charts in FIGS. 6 and 7 is performed by the control unit 12 controlling each constituent member of the three-dimensional shaping device 1 such as the supply unit 8 and the shaping table 9. FIG. 8 shows an example of forming the layer 501 that is a first layer of the layers 500. FIGS. 9 and 10 show an example of forming the second layer 502 of the layers 500. FIGS. 11 to 15 are conceptual diagrams showing bitmap data in which the layers 500 formed at the shaping table 9, the upper surface portion 10a of the table unit 10, and the liquid receiving unit 5 are divided into regions where the liquid can be ejected from the nozzle N of the head 3.

The three-dimensional shaping device 1 according to the present embodiment can form the layers 500 by moving the supply unit 8 in the X1 direction when forming each layer 500. Further, when each layer 500 is formed, the layers 500 can be formed by moving the supply unit 8 in the X1 direction in an odd-numbered layer 500 and moving the supply unit 8 in the X2 direction in an even-numbered layer 500. Here, FIG. 9 shows an example of forming the layers 500 by moving the supply unit 8 in the X1 direction not only when the layer 501 which is the odd-numbered layer 500 is formed but also when the layer 502 which is the even-numbered layer 500 is formed. FIG. 10 shows an example of forming the layers 500 by moving the supply unit 8 in the X1 direction when the odd-numbered layer 500 is formed and moving the supply unit 8 in the X2 direction when the even-numbered layer 500 is formed.

As shown in FIG. 6, first, in shaping data input processing in step S110, shaping data of a three-dimensional shaped object to be manufactured is input. An input source of the shaping data of the three-dimensional shaped object is not particularly limited, and the shaping data can be input to the three-dimensional shaping device 1 using the external device 20.

Next, in pre-shaping flushing processing in step S120, pre-shaping flushing is performed on the head 3. Here, the pre-shaping flushing is performed by moving the head 3 to the flushing position P2, which is the flushing position, facing the liquid receiving unit 5, and performing the pre-shaping flushing at the flushing position P2. The pre-shaping flushing processing in step S120 may be omitted.

Next, in layer forming processing in step S130, the shaping material is supplied from the shaping material supply units 2 to the shaping surface 9a of the shaping table 9, and the shaping material is compressed and leveled by the rollers 6 to form the layers 500. Here, the uppermost state in FIG. 8 represents a state where the supply unit 8 is moved in the X1 direction to form the first layer 501. The uppermost state in FIG. 9 represents a state where the supply unit 8 is moved in the X1 direction to form the second layer 502. In this way, when forming the layer 500 by moving the supply unit 8 in the X1 direction, the layer 500 is formed by

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supplying the shaping material from the shaping material supply unit 2A and compressing and leveling the shaping material by the roller 6A. On the other hand, as shown in the uppermost state in FIG. 10, when forming the layer 500 by moving the supply unit 8 in the X2 direction, the layer 500 is formed by supplying the shaping material from the shaping material supply unit 2B and compressing and leveling the shaping material by the roller 6B.

Next, in liquid ejection processing in step S140, the liquid containing the binder is ejected from the nozzle N of the head 3 to the shaping region P1 of the three-dimensional shaped object in the layer 500. A second state from the top in FIG. 8 represents a state where the liquid is ejected from the nozzle N of the head 3 to the shaping region P1 of the layer 501 while the supply unit 8 is moved in the X1 direction. A second state from the top in FIG. 9 represents a state where the liquid is ejected from the nozzle N of the head 3 to the shaping region P1 of the layer 501 while the supply unit 8 is moved in the X1 direction. In this way, when the layer 500 is formed by moving the supply unit 8 in the X1 direction, the liquid is ejected from the head 3A. On the other hand, as shown in a second state from the top in FIG. 10, when the layer 500 is formed by moving the supply unit 8 in the X2 direction, the liquid is ejected from the head 3B.

Next, in ultraviolet ray irradiation processing in step S150, the ultraviolet ray irradiation unit 4 irradiates the shaping region P1 of the three-dimensional shaped object in the layer 500 with ultraviolet rays. A third state from the top in FIG. 8 represents a state where the ultraviolet ray irradiation unit 4 irradiates the shaping region P1 of the three-dimensional shaped object in the layer 501 with ultraviolet rays while the supply unit 8 is moved in the X1 direction. A third state from the top in FIG. 9 represents a state where the ultraviolet ray irradiation unit 4 irradiates the shaping region P1 of the three-dimensional shaped object in the layer 502 with ultraviolet rays while the supply unit 8 is moved in the X1 direction. A third state from the top in FIG. 10 represents a state where the ultraviolet ray irradiation unit 4 irradiates the shaping region P1 of the three-dimensional shaped object in the layer 502 with ultraviolet rays while the supply unit 8 is moved in the X1 direction.

Next, in the flushing processing in step S160, flushing of the head 3 is performed. Here, as shown in the lowermost state in FIG. 8 and the lowermost state in FIG. 9, when the layer 500 is formed by moving the supply unit 8 in the X1 direction, flushing of the head 3A is performed at the flushing position P2 on a side facing the liquid receiving unit 5A. Following the flushing of the head 3A, the flushing of the head 3B can also be performed at the flushing position P2 on the side facing the liquid receiving unit 5A. On the other hand, as shown in the lowermost state in FIG. 10, when the layer 500 is formed by moving the supply unit 8 in the X2 direction, flushing of the head 3B is performed at the flushing position P2 on a side facing the liquid receiving unit 5B. Following the flushing of the head 3B, the flushing of the head 3A can also be performed at the flushing position P2 on the side facing the liquid receiving unit 5B.

Here, the flushing processing in step S160 will be described in detail with reference to FIG. 7. As shown in FIG. 7, when the flushing processing is started, first, in shaping data acquisition processing in step S1610, the control unit 12 acquires slice data corresponding to one layer 500 among the shaping data input in the shaping data input processing in step S110.

Next, in idle running time calculation processing in step S1620, an idle running time L is calculated by the control unit 12 based on the slice data which is bitmap data of one

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layer among the shaping data acquired in the shaping data acquisition processing in step S1610. Then, in first idle running time determination processing in step S1630, the control unit 12 determines whether the idle running time L is equal to or greater than a first threshold value. Instead of a configuration in which the idle running time L is calculated based on the slice data, a configuration in which an interval of a liquid ejection timing is measured by a timer (not shown) may be employed.

When it is determined in the first idle running time determination processing in step S1630 that the idle running time L is equal to or greater than the first threshold value, the process proceeds to second flushing signal generation processing in step S1640, and the control unit 12 generates a second flushing signal and executes a second flushing operation based on the second flushing signal. Then, in idle running time resetting processing in step S1670, information related to the idle running time L is reset, and the flushing processing in step S160 in FIG. 6 shown by the flow chart in FIG. 7 is completed. The control unit 12 includes a nonvolatile memory (not shown) including an EEROM or the like, and the information related to the idle running time L can be overwritten and stored in the nonvolatile memory.

When it is determined in the first idle running time determination processing in step S1630 that the idle running time L is less than the first threshold value, the process proceeds to second idle running time determination processing in step S1650, and the control unit 12 determines whether the idle running time L is equal to or greater than a second threshold value. Here, the second threshold value is, of course, a value smaller than the first threshold value. When it is determined in the second idle running time determination processing in step S1650 that the idle running time L is less than the second threshold value, the flushing processing in step S160 in FIG. 6 shown by the flowchart in FIG. 7 is completed. On the other hand, when it is determined in the second idle running time determination processing in step S1650 that the idle running time L is equal to or greater than the second threshold value, the process proceeds to first flushing signal generation processing in step S1660, and the control unit 12 generates a first flushing signal and executes the first flushing operation based on the first flushing signal. Then, in idle running time resetting processing in step S1670, information related to the idle running time L is reset, and the flushing processing in step S160 in FIG. 6 shown by the flow chart in FIG. 7 is completed.

That is, when the idle running time L is short and it is considered that almost no powder is mixed into the nozzle N, the flushing operation is temporarily suspended. As will be described later, when the idle running time L is long and it is considered that there is a high possibility that powder is mixed into the nozzle N, the flushing operation is executed. With completion of the flushing processing in step S160 in FIG. 6 shown by the flow chart in FIG. 7, the process proceeds to shaping data completion determination processing in step S170 in the flow chart in FIG. 6.

In the second state from the top in FIG. 8, the idle running time L is an idle running time L1. In the second state from the top in FIG. 9, the idle running time L is an idle running time L2A. In the second state from the top in FIG. 10, the idle running time L is an idle running time L2B. The idle running time L1 in the second state from the top in FIG. 8 is short, and is less than the first threshold value in the first idle running time determination processing in step S1630. Therefore, in such a case, it is determined that flushing does not have to be performed or weak flushing is sufficient, and

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after the first idle running time determination processing in step S1630 is completed, the process proceeds to the second idle running time determination processing in step S1650. In the present embodiment, for example, the control unit 12 determines that the idle running time L1 is equal to or greater than the second threshold value in the second idle running time determination processing in step S1650, and the process proceeds to the first flushing signal generation processing in step S1660. On the other hand, the idle running time L2A in the second state from the top in FIG. 9 and the idle running time L2B in the second state from the top in FIG. 10 are long and equal to or greater than the first threshold value. Therefore, in such a case, it is determined that strong flushing is necessary, and the process proceeds to the second flushing signal generation processing in step S1640.

Although the three-dimensional shaping device 1 according to the present embodiment includes the plurality of nozzles N in the head 3, the flushing operation of either the first flushing operation or the second flushing operation can be uniformly executed on all the nozzles N under the same condition regardless of the number of the nozzles N used for forming the structure S of the three-dimensional shaped object. In addition to such a flushing operation execution method, it is also possible to determine the idle running time L for each nozzle N and change whether to suspend the flushing operation, execute the first flushing operation, or execute the second flushing operation for each nozzle N. It should be noted that the second flushing operation corresponds to a stronger flushing operation than the first flushing operation, and how to set the second flushing operation to a stronger flushing operation than the first flushing operation will be described later.

Returning to the flow chart in FIG. 6, in the shaping data completion determination processing in step S170, the control unit 12 of the three-dimensional shaping device 1 determines whether formation of all the layers 500 based on the shaping data input in step S110 is completed. When it is determined that the formation of the layers 500 is not all completed, the process returns to the layer forming processing in step S130, and the next layer 500 is formed. On the other hand, when it is determined that the formation of the layers 500 is completely completed, the process proceeds to degreasing processing in step S180.

In the degreasing processing in step S180, a resin component of the structure S manufactured by repeating from the pre-shaping flushing processing in step S120 to the shaping data completion determination processing in step S170, such as a binder, is degreased using an external device or the like. Examples of the degreasing method include, but are not particularly limited to, a method of volatilizing the resin component by heating and a method of dissolving the resin component by immersing the structure S in a solvent. Note that the degreasing processing in step S180 may be omitted depending on a type of the three-dimensional shaped object to be manufactured, for example, when the three-dimensional shaped object formed of resin is manufactured.

Then, in sintering processing in step S190, the structure S degreased using the external device or the like in the degreasing processing in step S180 is heated to sinter the shaping material. Even when the resin component such as a binder of the structure S remains even after the degreasing processing in step S180 is performed, the resin component is removed as the sintering processing in step S190 is executed. Then, along with completion of the sintering processing in step S190, a method for manufacturing the

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three-dimensional shaped object according to the present example is completed. As in the degreasing processing in step S180, the sintering processing in step S190 may be omitted depending on the type of the three-dimensional shaped object to be manufactured or the like.

Hereinafter, a flushing operation method for each nozzle N according to the structure S of the three-dimensional shaped object to be formed will be described with reference to FIGS. 11 to 15. FIGS. 11 to 15 are conceptual diagrams showing bitmap data in which the layers 500, the upper surface portion 10a, and the liquid receiving unit 5 are divided into regions where the liquid can be ejected from the nozzle N of the head 3 as described above. In other words, the layers 500, the upper surface portion 10a, and the liquid receiving unit 5 are divided into areas for each ejection unit of the liquid from the nozzle N. FIGS. 11 to 15 show a liquid ejection area and a liquid non-ejection area when the liquid is ejected while the head 3 is moved in the X1 direction. A black area in the drawing is the liquid ejection area and corresponds to ejection data D1, and a white area in the drawing is the liquid non-ejection area and corresponds to NULL data D0.

Here, shaping region data Da corresponding to the ejection of the liquid in the layers 500 including the shaping region P1 includes the ejection data D1 for ejecting the liquid corresponding to the formation of the structure S and the NULL data D0 for not ejecting the liquid, which are divided for each area. That is, the liquid is ejected to a region of the ejection data D1, and the liquid is not ejected to a region of the NULL data D0. Idle running region data Db corresponding to the ejection of the liquid on the upper surface portion 10a that does not include the shaping region P1 includes only the NULL data D0 divided for each area. That is, the liquid is not ejected to the upper surface portion 10a. Flushing data Dc corresponding to the ejection of the liquid in the liquid receiving unit 5 includes the ejection data D1 corresponding to the flushing area on the liquid receiving unit 5 and the NULL data D0 not corresponding to the flushing area on the liquid receiving unit 5, which are divided for each area. That is, the liquid is ejected to a region of the ejection data D1, and the liquid is not ejected to a region of the NULL data D0. Here, the shaping region data Da, the idle running region data Db, and the flushing data Dc are generated by the control unit 12 based on the shaping data input in the shaping data input processing in step S110.

First, a case shown in FIG. 11 will be described. FIG. 11 shows a case where ejection is performed on an area corresponding to the structure S by using each nozzle N of the head 3, and the idle running time L for any of the nozzle N is an idle running time La equal to or greater than the first threshold value. In such a case, the flushing operation is performed at the liquid receiving unit 5 on any of the nozzles N. In the case shown in FIG. 11, it is determined that the idle running time L is equal to or greater than the first threshold value in the first idle running time determination processing in step S1630, and the second flushing operation corresponding to the second flushing signal generation processing in step S1640 is executed at the liquid receiving unit 5.

Next, a case shown in FIG. 12 will be described. Similar to FIG. 11, FIG. 12 shows a case where ejection is performed on an area corresponding to the structure S by using each nozzle N of the head 3, and the idle running time L for any of the nozzles N is an idle running time Lb less than the first threshold value and less than the second threshold value. In such a case, the flushing operation is not performed at the liquid receiving unit 5 on any of the nozzles N.

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Next, a case shown in FIG. 13 will be described. FIG. 13 shows a case where, unlike FIGS. 11 and 12, ejection is performed on an area corresponding to the structure S by using only the nozzles N in a region R1 among the nozzles N in the region R1, a region R2, and a region R3, the idle running time L in the nozzles N of the region R1 is less than the first threshold value and less than the second threshold value, and the idle running time L in the nozzles N of the region R2 and the region R3 is equal to or greater than the first threshold value. Each of the region R1, the region R2, and the region R3 corresponds to a strip-shaped region extending in the X direction in the drawing in the layers 500, the upper surface portion 10a, and the liquid receiving unit 5. Here, in the case shown in FIG. 13, the flushing operation is not performed at the liquid receiving unit 5 on the nozzles N in the region R1, and the second flushing operation corresponding to the second flushing signal generation processing in step S1640 is executed at the liquid receiving unit 5 on the nozzles N in the region R2 and the region R3.

Next, a case shown in FIG. 14 will be described. Similarly to FIGS. 11 and 12, FIG. 14 shows a case where ejection is performed on an area corresponding to the structure S by using the nozzles N in all the regions among the nozzles N in the region R1, the region R2, and the region R3, the idle running time L in the nozzles N in the region R1 is less than the first threshold value and less than the second threshold value, and the idle running time L in the nozzles N in the region R2 and the region R3 is less than the first threshold value and equal to or greater than the second threshold value. In such a case, the flushing operation is not performed at the liquid receiving unit 5 on the nozzles N in the region R1, and the first flushing operation corresponding to the first flushing signal generating processing in step S1660 is executed at the liquid receiving unit 5 on the nozzles N in the region R2 and the region R3. As shown in FIG. 14, when the first flushing operation is executed in the region R2 and the region R3, the flushing areas corresponding to the ejection data D1 are arranged alternately with respect to the region on the liquid receiving unit 5. By arranging the flushing area in this manner, the liquid can be efficiently absorbed by an absorber (not shown) provided in the liquid receiving unit 5, and the liquid can be prevented from spilling out from the liquid receiving unit 5. In other words, by changing the ejection position (flushing area) and the number of times of ejection of the liquid without changing a droplet size of the liquid, the liquid can be efficiently absorbed by the absorber (not shown) provided in the liquid receiving unit 5. The flushing areas in the second flushing operation shown by the liquid receiving unit 5 in FIG. 11 and the region R2 and the region R3 in FIG. 13 and FIG. 15 described later are also evenly arranged as shown in the drawing.

Next, a case shown in FIG. 15 will be described. Similar to FIG. 13, FIG. 15 shows a case where ejection is performed on an area corresponding to the structure S by using only the nozzles N in the region R1 among the nozzles N in the region R1, the region R2, and the region R3, the idle running time L in the nozzle N in the region R1 is less than the first threshold value and equal to or greater than the second threshold value, and the idle running time L in the nozzles N in the region R2 and the region R3 is equal to or greater than the first threshold value. In such a case, the first flushing operation corresponding to the first flushing signal generation processing in step S1660 is executed at the liquid receiving unit 5 on the nozzles N in the region R1, and the second flushing operation corresponding to the second flushing signal generation processing in step S1640 is executed on the liquid receiving unit 5 on the nozzles N in the region

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R2 and the region R3. As shown in FIG. 15, when the first flushing operation is executed in the region R1, the flushing areas corresponding to the ejection data D1 are arranged alternately with respect to the region on the liquid receiving unit 5.

In summary, in the three-dimensional shaping device 1 according to the present embodiment, the control unit 12 changes the execution condition of the flushing operation depending on whether the idle running time L during which the head 3 is moved without ejecting the liquid is less than the threshold value or equal to or greater than the threshold value. For this reason, for example, when the idle running time L becomes long and the time during which the head 3 is moved on the layers 500 which are powder layers in an idle running state becomes long, it is possible to discharge the liquid in the head 3 under a condition where the flushing effect is high. Therefore, it is possible to prevent the powder mixed into the nozzle N from not being removed even though the flushing operation is performed. The “idle running time” in the present embodiment indicates a time during which the head 3 is moved on the powder layer or a time during which the head 3 stays on the powder layer at least in a state where no droplet is ejected from the nozzle N of the head 3.

As described above, the shaping area data Da, the idle running region data Db, and the flushing data Dc are generated by the control unit 12 based on the shaping data input in the shaping data input processing in step S110. That is, the control unit 12 generates the flushing data Dc at the flushing position P2 based on the shaping data of the three-dimensional shaped object, and controls the head 3 to execute the flushing operation based on the flushing data Dc. In this way, the flushing operation can be made appropriate by generating the flushing data Dc based on the shaping data. For example, by ejecting the liquid so as to be uniform with respect to the liquid receiving unit 5 as shown by the region R2 and the region R3 in FIG. 14 and the liquid receiving unit 5 which is the flushing position P2 in the region R1 in FIG. 15, the liquid can be efficiently absorbed by the absorber accommodated in the liquid receiving unit 5, and spilling of the liquid from the liquid receiving unit 5 due to the flushing operation can be prevented.

The control unit 12 does not execute the flushing operation when the idle running time L is less than the second threshold value as the threshold value, and executes the flushing operation when the idle running time L is equal to or greater than the first threshold value or the second threshold value as the threshold value. That is, the control unit 12 does not execute the flushing operation when the idle running time L is less than the threshold value. Therefore, the three-dimensional shaping device 1 according to the present embodiment can effectively prevent the liquid from being wasted due to the flushing operation being performed even when the flushing operation is unnecessary.

The three-dimensional shaping device 1 according to the present embodiment will be described from another viewpoint. The control unit 12 controls the head 3 to execute the first flushing operation when the idle running time L is less than the first threshold value, and controls the head 3 to execute the second flushing operation under a flushing condition different from that of the first flushing operation when the idle running time L is equal to or greater than the first threshold value. For this reason, for example, when the idle running time L during which the head 3 is moved on the layers 500 which are powder layers in an idle running state becomes long, it is possible to discharge the liquid in the head 3 under the condition where the flushing effect is high.

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Therefore, it is possible to prevent the powder mixed into the nozzle N from not being removed even though the flushing operation is performed.

Here, the three-dimensional shaping device 1 according to the present embodiment resets the measurement start timing of the idle running time L with the ejection of the liquid from the head 3 accompanying the formation of the structure S under the control of the control unit 12. Further, as shown in the flow chart in FIG. 7, every time the control unit 12 executes the second flushing signal generation processing in step S1640 and the first flushing signal generation processing in step S1660, the control unit 12 resets the information related to the idle running time L in the idle running time resetting processing in step S1670. That is, the control unit 12 resets the measurement start timing of the idle running time L every time the flushing operation is executed. In other words, depending on a use timing of the nozzle N corresponding to a shape, size, or the like of the structure S to be formed, the measurement start timing of the idle running time L is reset every time a plurality of layers 500 are formed. Therefore, since the number of times of the flushing operation can be reduced as compared with a case where the measurement start timing of the idle running time L is reset each time one layer of the layers 500 is formed, it is possible to effectively discharge the powder mixed into the nozzle N while effectively preventing waste of the liquid. The “measurement start timing” is not limited to a timing at which measurement is actually started using a timer or the like, and includes a start timing of a time calculated based on data by the control unit 12.

The three-dimensional shaping device 1 according to the present embodiment can also reset the measurement start timing of the idle running time L every time one layer 500 is formed under the control of the control unit 12. Specifically, in the flow chart in FIG. 7, when it is determined that the idle running time L is less than the second threshold value in the second idle running time determination processing in step S1650, the control unit 12 can controls such that the process proceeds to the idle running time resetting processing in step S1670 and then the flushing processing in step S160 is completed. By performing such control, it is possible to effectively discharge the powder mixed into the nozzle N with simple control.

The three-dimensional shaping device 1 according to the present embodiment executes the second flushing operation when the idle running time L is equal to or greater than the first threshold value, executes the first flushing operation when the idle running time L is less than the first threshold value and equal to or greater than the second threshold value, and does not execute the flushing operation when the idle running time L is less than the second threshold value. However, the flushing operation may be executed when the idle running time L is equal to or greater than one threshold value, and the flushing operation may not be executed when the idle running time L is less than the threshold value. Further, the second flushing operation may be executed when the idle running time L is equal to or greater than one threshold value, and the first flushing operation may be executed when the idle running time L is less than the threshold value. Further, a plurality of threshold values may be set, and in addition to the first flushing operation and the second flushing operation, a flushing operation may be performed under flushing conditions different from those of the first flushing operation and the second flushing operation. Instead of the configuration in which the execution condition of the flushing operation is changed depending on whether the idle running time L is less than the threshold

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value or equal to or greater than the threshold value, a configuration may be used in which the execution condition of the flushing operation is changed depending on whether the idle running time L is equal to or less than the threshold value or exceeds the threshold value.

The three-dimensional shaping device 1 according to the present embodiment can make a frequency of a waveform input to the head 3 in the second flushing operation higher than a frequency of a waveform input to the head 3 in the first flushing operation. When the frequency of the waveform input to the head 3 is increased, the powder mixed into the nozzle N can be effectively discharged, and therefore, the three-dimensional shaping device 1 according to the present embodiment can effectively discharge the powder mixed into the nozzle N even when the idle running time L becomes long.

Further, the three-dimensional shaping device 1 according to the present embodiment can make a voltage to be applied to the head 3 in the second flushing operation higher than a voltage to be applied to the head 3 in the first flushing operation. When the voltage applied to the head 3 is increased, the powder mixed into the nozzle N can be effectively discharged, and therefore, the three-dimensional shaping device 1 according to the present embodiment can effectively discharge the powder mixed into the nozzle N even when the idle running time L becomes long.

Further, the three-dimensional shaping device 1 according to the present embodiment can make an ejection speed of the liquid from the head 3 in the second flushing operation faster than an ejection speed of the liquid from the head 3 in the first flushing operation by adjusting the voltage to be applied to the head 3 and the waveform when the voltage is applied to the head 3. When the ejection speed of the liquid from the head 3 is increased, the powder mixed into the nozzle N can be effectively discharged, and therefore, the three-dimensional shaping device 1 according to the present embodiment can effectively discharge the powder mixed into the nozzle N even when the idle running time L becomes long.

Further, the three-dimensional shaping device 1 according to the present embodiment can make a droplet size of the liquid ejected from the head 3 in the second flushing operation larger than a droplet size of the liquid ejected from the head 3 in the first flushing operation by adjusting the voltage to be applied to the head 3 and the waveform when the voltage is applied to the head 3. When an ejection amount of the liquid from the head 3 is increased, the powder mixed into the nozzle N can be effectively discharged, and therefore, the three-dimensional shaping device 1 according to the present embodiment can effectively discharge the powder mixed into the nozzle N even when the idle running time L becomes long.

In the flow chart in FIG. 6, the pre-shaping flushing processing in S160 is executed. The flushing operation here is the first flushing operation. In addition to the pre-shaping flushing processing in step S160 or instead of the pre-shaping flushing processing in step S160, post-shaping flushing processing may be executed by the first flushing operation after the shaping data completion determination processing in step S170 and before the degreasing processing in step S180. That is, the control unit 12 may control such that the first flushing operation is executed at least when the operation of the three-dimensional shaping device 1 is started or when the operation of the three-dimensional shaping device 1 is completed. This is because the formation of the structure S can be started in a state where the powder is not mixed into the nozzle N by executing the first flushing operation at least when the operation of the three-dimensional

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sional shaping device 1 is started or when the operation of the three-dimensional shaping device 1 is completed. Therefore, when the three-dimensional shaping device 1 is operated, it is possible to prevent occurrence of a liquid ejection failure due to the powder mixed into the nozzle N. By setting the flushing operation when the operation of the three-dimensional shaping device 1 is started and when the operation of the three-dimensional shaping device 1 is completed as the first flushing operation instead of the second flushing operation, wasteful consumption of the liquid can be prevented.

In the three-dimensional shaping device 1 according to the present embodiment, the control unit 12 controls the flushing operation while keeping a moving speed of the head 3 constant, and the control unit 12 may change the moving speed of the head 3 according to shaping quality of the shaping data. For example, when the moving speed of the head 3 is decreased in order to increase the accuracy of a shaped object, a time for the head 3 to move on the powder layer is increased. Therefore, even when the idle running time varies depending on the moving speed of the head 3, the control unit 12 controls the flushing operation, so that the powder mixed into the nozzle N can be effectively discharged.

The present disclosure is not limited to the embodiments described above, and can be implemented in various configurations without departing from the scope of the disclosure. For example, in order to solve some or all of problems described above, or to achieve some or all of effects described above, technical characteristics in the embodiments corresponding to the technical characteristics in each embodiment described in the summary of the disclosure can be replaced or combined as appropriate. In addition, when the technical characteristics are not described as essential in the present description, the technical characteristics can be appropriately deleted.

What is claimed is:

1. A three-dimensional shaping device, comprising:

a shaping table;

a layer forming unit configured to form a powder layer on the shaping table;

a head configured to eject a liquid containing a binder from a nozzle to a shaping region of a three-dimensional shaped object on the powder layer; and

a control unit configured to control movement of the head with respect to the shaping table and driving of the head by applying a voltage, and to control the head to execute a flushing operation at a flushing position which is a position different from the shaping region, wherein

the control unit changes an execution condition of the flushing operation depending on whether an idle running time during which the head is moved without ejecting the liquid is less than a threshold value or the idle running time during which the head is moved without ejecting the liquid is equal to or greater than the threshold value.

2. The three-dimensional shaping device according to claim 1, wherein

the control unit is configured to generate flushing data related to ejection of the liquid at the flushing position based on shaping data of the three-dimensional shaped object, and control the head to execute the flushing operation based on the flushing data.

3. The three-dimensional shaping device according to claim 1, wherein

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the control unit does not execute the flushing operation when the idle running time is less than the threshold value, and executes the flushing operation when the idle running time is equal to or greater than the threshold value.

4. The three-dimensional shaping device according to claim 1, wherein the flushing operation comprises a first flushing operation and a second flushing operation, and the control unit controls the head to execute the first flushing operation when the idle running time is less than the threshold value, and controls the head to execute the second flushing operation when the idle running time is equal to or greater than the threshold value.

5. The three-dimensional shaping device according to claim 4, wherein
a frequency of a waveform input to the head in the second flushing operation is higher than a frequency of a waveform input to the head in the first flushing operation.

6. The three-dimensional shaping device according to claim 4, wherein
a voltage to be applied to the head in the second flushing operation is higher than a voltage to be applied to the head in the first flushing operation.

7. The three-dimensional shaping device according to claim 4, wherein
an ejection speed of the liquid from the head in the second flushing operation is faster than an ejection speed of the liquid from the head in the first flushing operation.

8. The three-dimensional shaping device according to claim 4, wherein
a droplet size of the liquid ejected from the head in the second flushing operation is larger than a droplet size of the liquid ejected from the head in the first flushing operation.

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9. The three-dimensional shaping device according to claim 4, wherein

the control unit controls to execute the first flushing operation at least when the operation of the three-dimensional shaping device is started or when the operation of the three-dimensional shaping device is completed.

10. The three-dimensional shaping device according to claim 1, wherein

the control unit resets a measurement start timing of the idle running time every time one powder layer is formed.

11. The three-dimensional shaping device according to claim 1, wherein

the control unit resets a measurement start timing of the idle running time every time the flushing operation is executed.

12. The three-dimensional shaping device according to claim 1, wherein

the head includes a pressure chamber communicating with the nozzle, a supply path configured to supply the liquid to the pressure chamber, and a circulation path into which the liquid from the pressure chamber flows for circulation.

13. The three-dimensional shaping device according to claim 12, wherein

the control unit is configured to control such that q_2/q_1 is equal to or greater than 0.05 and equal to or less than 20, where q_1 is a flow rate of the liquid flowing into the circulation path per unit time, and q_2 is a maximum flow rate of the liquid ejected from the nozzle per unit time.

14. The three-dimensional shaping device according to claim 13, wherein
 q_2/q_1 is 0.05.

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